

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 521 — Past Questions & Materials
by Renance

Course: GET 521
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 500

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Paper 1 of 2 — GET 521 TEST QUESTION.

Level 500 • Second Semester • PastQuestion

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Paper 2 of 2 — GET 521

Level 500 • Second Semester • Handout

GET 521

ENGINEERING MANAGEMENT

LECTURE NOTE

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GET 521: ENGINEERING MANAGEMENT 2 UNITS

Course Outline

1. Basic Definition of Engineering Management and Economics
2. Basic Concepts for Economic Assessment of Engineering Projects: Accounting methods; marginal analysis; cash flow and time value of money; taxation and depreciation; discounted cash flow analysis techniques; cost of capital; inflation;
3. Sensitivity and Risk Analysis; replacement decision; analysis of R and D; on-going as well as new investment opportunities;
4. Engineering and Public Sector Decision Making; qualitative considerations and multiple criteria.
5. Engineering Business: type, structure, goals and functions of organizations;
6. Management of Engineer to Manage Transition; R&D management; project selection and Project management; advanced manufacturing systems management; capital investment policies; technological collaborations: sub-contracting, consultancies, joint ventures and linkage programmes; management of change and innovation.

Course Objectives

This course is designed to meet the continuing development needs of Engineers for evaluating different

alternatives of engineering projects. Hence, the objectives are;

i. To enable Engineering students / Engineers learn the basic fundamentals of management and economics

as it applies to engineering

ii. To understand basic accounting and business principles in order to communicate with accountants and

managers

iii. To enable students know how to undertake R&D

iv. To enable students know how to manage project and resources in more efficient manner

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MODULE 1

INTRODUCTION

1.1 Basic Definition of Engineering Management

Engineering management is the application of the practice of management to engineering.

Management is a process (planning, organising, staffing, directing and controlling) of getting the work or the task done that is required for achieving the goals of an organisation in an efficient and effective manner.

Engineering is the practice of using or applying natural science, mathematics, and the scientific method to

solve problems, increase efficiency and productivity, and improve systems. Modern engineering consists of

different disciplines such as Mechanical, Electrical/Electronics, Chemical, Petroleum, Civil, Marine, etc, and

each discipline comprises many subfields.

Engineering management is a career that brings together the technological problem-solving ability of

engineering and the organizational, administrative, legal and planning abilities of management in order to

oversee the operational performance of complex engineering-driven enterprises. Engineering management

covers the gap between engineering and business management, namely the combination of technical and

economic decision-making with analytical skills, optimization capabilities, and technical product development.

Increasingly, all industries are becoming more data-driven and thus are basing their business decisions on all

the digital data collected across the supply chain. Because of the speed of innovation in this new era,

engineers are being increasingly brought to the decision table to analyze and interpret the massive amounts of

data to help make fast, robust economic decisions, and to apply engineering management skills to better lead

engineering teams to exploit new opportunities and develop smarter products, platforms, and services. With

the rise of Industry 4.0, where technologies combine human, machines, and data, a modernized form of

engineering management has arisen. In this new era, engineering management must have their decision

making supercharged by data and artificial intelligence (AI).

At its fundamental core, the field of engineering management combines technical engineering expertise,

human leadership, specific business management concepts, and advanced technologies. Today's engineering

managers must be skilled Industry 4.0 practitioners and leaders who can guide diverse engineering teams and

oversee complex service or product development that always leverage and consider the role of data. Their

responsibilities include safety, sustainability, resource allocation, team building, project planning, budgeting, and product development linked with economic and financial outcomes. They must also oversee execution in terms of technology deployment.

1.2 Important of Engineering Management

Effective engineering management matters enormously. It influences an organization's financial outcomes

(revenue growth, profitability and operational efficiency) and impacts engineering facilities, largescale

projects, clients, workers, and the public. The focus is on effective implementation of best practices to lead

engineering teams with knowledge spanning the gap between business management and engineering.

Engineering managers serve as the interface between the business and technical sides of a project, translating

business-oriented goals and needs into actionable engineering strategies and projects. They direct teams to

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ensure they stay motivated, focused, and efficient. Their joint knowledge of certain key aspects of business

management and technical engineering concepts allows them to proactively make balanced decisions that

will mitigate project risks, efficiently allocate resources, and consistently deliver high-quality products

within the project budget and schedule.

1.3 Engineering Economics

Engineering Economics is a subset of economics concerned with the use and application of economic

principles in the analysis of engineering decisions. As a discipline, it is focused on the branch of economics

known as microeconomics in that it studies the behavior of individuals and firms in making decisions

regarding the allocation of limited resources. Thus, it focuses on the decision making process, its context and

environment. It is pragmatic by nature, integrating economic theory with engineering practice.

But, it is also a simplified application of microeconomic theory in that it avoids a number of microeconomic

concepts such as price determination, competition and demand/supply. As a discipline though, it is closely

related to others such as statistics, mathematics and cost accounting.

Engineers seek solutions to problems, and the economic viability of each potential solution is normally

considered along with the technical aspects. Fundamentally, engineering economics involves formulating, estimating, and evaluating the economic outcomes when alternatives to accomplish a defined purpose are available.

Considering the time value of money is central to most engineering economic analyses. Cash flows are discounted using an interest rate. For each problem, there are usually many possible alternatives. The opportunity cost of making one choice over another must also be considered. Costs as well as revenues are considered, for each alternative, for an analysis period that is either a fixed number of years or the estimated life of the project. The salvage value is often important, and is either the net cost or revenue for decommissioning the project.

Some other topics that may be addressed in engineering economics are inflation, uncertainty, replacements, depreciation, resource depletion, taxes, accounting, cost estimations, and capital budgeting. All these topics are primary skills and knowledge areas in the field of cost engineering. Since engineering is an important part of the manufacturing sector of the economy, engineering economics is an important part of industrial or business economics.

1.4 Microeconomics

Microeconomics is a branch of economics that studies the behavior of individuals and firms in making decisions regarding the allocation of scarce resources and the interactions among these individuals and firms. One goal of microeconomics is to analyze the market mechanisms that establish relative prices among goods and services and allocate limited resources among alternative uses. Microeconomics shows conditions under which free markets lead to desirable allocations. It also analyzes market failure, where markets fail to produce efficient results. Macroeconomics involves the sum total of economic activity, dealing with the issues of growth, inflation, and unemployment and with national policies relating to these issues.

1.5 Why Do Engineers Need to Learn About Economics

Ages ago, the most significant barriers to engineers were technological. The things that engineers wanted to do, they simply did not yet know how to do, or hadn't yet developed the tools to do. There are certainly many

more challenges like this which face present-day engineers. However, we have reached the point in
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engineering where it is no longer possible, in most cases, simply to design and build things for the sake of

simply designing and building them. Natural resources (from which we must build things) are becoming

scarcer and more expensive. We are much more aware of negative side-effects of engineering innovations

(such as air pollution from automobiles) than ever before.

For these reasons, engineers are tasked more and more to place their project ideas within the larger

framework of the environment within the planet, country, or region. Engineers must ask themselves if a

particular project will offer some net benefit to the people who will be affected by the project, after considering its inherent benefits, plus any negative side-effects, along with the cost of consuming natural

resources, both in the price that must be paid for them and the realization that once they are used for that

project, they will no longer be available for any other project(s).

Simply put, engineers must decide if the benefits of a project exceed its costs, and must make this comparison in a unified framework. The framework within which to make this comparison is the field of

engineering economics, which strives to answer exactly these questions, and perhaps more. It should be clear

from this discussion that consideration of economic factors is as important as regard for the physical laws

and science that determine what can be accomplished with engineering. The following figure shows how

engineering is composed of physical and economic components:

Figure 1 : Physical and Economic Components of an Engineering System

Figure 1 shows how engineering is composed of physical and economic components.

1. Physical Environment: Engineers produce products and services depending on physical laws (e.g. Ohm's law; Newton's law).

Physical efficiency takes the form:

system output(s)

Physical (efficiency) = -----

system input(s)

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2. Economic Environment: Much less of a quantitative nature is known about economic environments-

this is due to economics being involved with the actions of people, and the structure of organizations.

Satisfaction of the physical and economic environments is linked through production and construction processes. Engineers need to manipulate systems to achieve a balance in attributes in both the physical and economic environments, and within the bounds of limited resources. In the final

evaluation of most ventures, economic efficiency takes precedence over physical efficiency because

projects cannot be approved, regardless of their physical efficiency, if there is no conceived demand

for them amongst the public, if they are economically infeasible, or if they do not constitute the "wisest" use of those resources which they require.

There are numerous examples of engineering systems that have physical design but little economic worth (i.e

it may simply be too expensive). Consider a proposal to purify all of the water used by a large city by boiling

it and collecting it again through condensation. This type of experiment is done in junior physical science

labs every day, but at the scale required by a large city, is simply too costly.

1.6 Role of Uncertainty in Engineering

When conducting engineering economic analyses, it will be assumed at first, for simplicity, that benefits,

costs, and physical quantities will be known with a high degree of confidence. This degree of confidence is

sometimes called assumed certainty. In virtually all situations, however, there is some doubt as to the

ultimate values of various quantities. Both risk and uncertainty in decision-making activities are caused by a

lack of precise knowledge regarding future conditions, technological developments, synergies among funded

projects, etc. Decisions under risk are decisions in which the analyst models the decision problem in terms of

assumed possible future outcomes, or scenarios, whose probabilities of occurrence can be estimated. Of

course, this type of analysis requires an understanding of the field of probability. Decisions under uncertainty, by contrast, are decision problems characterized by several unknown futures for which probabilities of occurrence cannot be estimated. Other less objective means exist for the analysis of such

problems.

For the purposes of this brief tutorial, we cannot delve further into the analytical extensions required to

accommodate risk or uncertainty in the decision process. We must recognize that these things exist, however,

and be careful about reaching strong conclusions based on data which might be susceptible to these. Because

engineering is concerned with actions to be taken in the future, an important part of the engineering process

is improving the certainty of decisions with respect to satisfying the objectives of engineering applications.

1.7 The Engineering Process

Engineering activities dealing with elements of the physical environment take place to meet human needs

that arise in an economic setting. The engineering process employed from the time a particular need is

recognized until it is satisfied may be divided into a number of phases:

i. Determination of Objectives: This step involves finding out what people need and want that can be supplied by engineering. People's wants may arise from logical considerations, emotional drives, or a combination of the two.

ii. Identification of Strategic Factors: The factors that stand in the way of attaining objectives are known as limiting factors. Once the limiting factors have been identified, they are examined to locate strategic factors -- those factors which can be altered to remove limitations restricting the success of an undertaking. A woman who wants to empty the water from her swimming pool might be faced with the limiting factor that she only has a bucket to do the job with, and this would require far greater time and physical exertion than she has at her disposal. A strategic factor developed in

6 response to this limitation would be the procurement of some sort of pumping device which could do the job much more quickly, with almost no physical effort on the part of the woman.

iii. Determination of means (engineering proposals): This step involves discovering what means exist to alter strategic factors in order to overcome limiting factors. In the previous example, one means was to buy (or rent) a pump. Of course, if the woman had a garden hose, she might have been able to siphon the water out of the pump. In other engineering applications, it may be necessary to fabricate the means to solve problems from scratch.

iv. Evaluation of Engineering Proposals: It is usually possible to accomplish the same result with a variety of means. Once these means have been described fully, in the form of project proposals, economic analysis can be employed to determine which among them, if any, is the best means for solving the problem at hand.

v. Assistance in Decision Making: It is commonplace for the final decision-making responsibility to fall on the head(s) of someone other than the engineer(s). The person(s) so charged, however, may not be

sufficiently knowledgeable about the technical aspects of a proposal to determine its relevant worth

compared to other means. The engineer can help to bridge this gap.

1.8 Engineering Economic Studies

The four key steps in planning an economic study are:

1. Creative Step: People with vision and initiative adopt the premise that better opportunities exist than

are known to them. This leads to research, exploration, and investigation of potential opportunities.

2. Definition Step: System alternatives are synthesised with economic requirements and physical requirements, and enumerated with respect to inputs/outputs.

3. Conversion Step: The attributes of system alternatives are converted to a common measure so that

systems can be compared. Future cash flows are assigned to each alternative, consisting of the time-

value of money.

4. Decision Step: Qualitative and quantitative inputs and outputs to/from each system form the basis for

system comparison and decision making. Decisions among system alternatives should be made on the

basis of their differences. For a small number of real world systems there will be complete knowledge. All facts/information and their relationships, judgements and predictive behavior become

a certainty. For most systems, however, even after all of the data that can be bought to bear on it has

been considered, some areas of uncertainty are likely to remain. If a decision must be made, these areas of uncertainty must be bridged by consideration of non-quantitative data/information, such as

common sense, judgement and so forth. Decisions among system alternatives should be made on the

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uncertainty must be bridged by consideration of non-quantitative data/information, such as common

sense, judgement and so forth.

Examples :

1. Infrastructure expenditure decision
2. Replace versus repair decisions
3. Selection of inspection method

4. Selection of a replacement for an equipment

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1.8.1 Fundamental Economic Concepts

Economics deals with the behavior of people, and as such, economic concepts are usually qualitative in nature, and not universal in application.

Utility: Utility is the power of a good or service to satisfy human needs.

Value: it designates the worth that a person attaches to an object or service. It is a measure or appraisal of utility in some medium of exchange. It is not the same as cost or price.

1.9 Consumer and Producer Goods

i. Consumer goods: Consumer goods are the goods and services that directly satisfy human wants. Basic

human needs of food, clothing and shelter. In commercial advertisements, emphasis is given to senses not reasoning. For example, TV, shoes, houses.

ii. Producer goods: Producer goods are the goods and services that satisfy human wants indirectly as a part of the production or construction process. For example, factory equipment, industrial chemicals and materials. The utility stems for their means to get to an end.

1.10 Economy of Exchange

1. Occurs when utilities are exchanged by two or more people.
2. It is possible because consumer utilities are evaluated subjectively.
3. Represents mutual benefit in exchange.
4. Persuasion in exchange. Salesperson.

1.11 Classification of Cost

A key objective in engineering applications is the satisfaction of human needs, which will nearly always

imply a cost. Economic analyses may be based on a number of cost classifications:

1. First (or Initial) Cost: Cost to get activity started such as property improvement, transportation, installation, and initial expenditures.
2. Operation and Maintenance Cost: They are experienced continually over the useful life of the activity.
3. Fixed Cost: Fixed costs arise from making preparations for the future, and includes costs associated with ongoing activities throughout the operational life-time of that concern. Fixed costs are relatively constant; they are decoupled from the system input/output, for example.
4. Variable Cost: Variable costs are related to the level of operational activity (e.g. the cost of fuel for construction equipment will be a function of the number of days of use).

5. Incremental or Marginal Cost: Incremental (or marginal) cost is the additional expense that will be

incurred from increased output in one or more system units (i.e. production increase). It is determined

from the variable cost.

6. Sunk Cost: It cannot be recovered or altered by future actions. Usually this cost is not a part of engineering economic analysis.

7. Life-Cycle Cost: This is cost for the entire life-cycle of a product, and includes feasibility, design, construction, operation and disposal costs.

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MODULE 2

BASIC CONCEPTS FOR ECONOMIC ASSESSMENT OF ENGINEERING PROJECTS

2.1 Accounting methods

Accounting methods refer to the set of rules a business follows to keep track of financial transactions and

financial records. Its main objective is to provide an accurate overview of an organization's expenses and

profits.

2.1.1 Importance of Using the Right Account Method

Here are some of the reasons why choosing the right method are:

i. Determine financial health: Each accounting method offers various levels of perceived profitability in

the short-term and long-term. Some methods give a status of your business's current health based on

current true income, while others provide a long-term overview of a business' finances.

ii. Taxation purposes: Based on company type, you may be required to report finances using a specific

accounting method.

iii. Financing purposes: The right accounting method can improve your chances when looking for debt

financing and when being evaluated by investors or other financial institutions.

iv. Growth forecasting: The right accounting method will make it easier for you to forecast the future

growth of your business based on your company's size.

2.2 Types of Accounting Methods

An accounting method refers to the rules a company follows in reporting revenues and expenses. The two

primary methods of accounting are cash accounting (generally used by individuals) and accrual accounting

(generally used by companies); additionally the third method is the Modified Cash-Basis Accounting method. Hence, the methods are;

i. Cash Accounting method

ii. Accrual Accounting method

iii. Modified Cash-Basis Accounting method

i. Cash Accounting method: Cash-basis accounting requires businesses to record cash, expenses and

income. Keep in mind income is recorded once received and expenses are paid. Cash accounting is also

called cash-basis accounting; and may be contrasted with accrual accounting, which recognizes income at

the time the revenue is earned and records expenses when liabilities are incurred regardless of when cash is

actually received or paid.

If you have a long-term project, you won't be able to record transactions until it is paid. Similarly, taxable

income and expenses are reported in the year in which you get the compensation or pay the expense. While

this method gives an accurate overview of a business' perceived income, it does not allow you to track loans,

liabilities and inventory. It also does not give an accurate representation of your business's status because

you won't be able to see unpaid transactions and expenses until the compensation is sent. Still, this method

may be ideal for small businesses that want an accurate overview of their financial situation.

Use With: Single-Entry Accounting: Cash-basis accounting is used alongside single-entry accounting because they are both the simplest accounting methods. Incoming receipts and outgoing expenses are only

recorded once.

ii. Accrual Accounting method

Accrual accounting is a more complex accounting method that requires you to record incoming revenue and

expenses—even if payment has not been made. This means expenses are recorded once the bill is received

and income for a long-term contract is recorded when the deal is closed.

Unlike the straightforward cash-based accounting, accrual accounting also considers accounts payable,

liabilities, assets and inventory.

From a business perspective, this method allows a business to record all incoming revenue and gives a better

understanding of their profitability. It also makes it easy to create financial forecasts and estimate budgets.

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Since all incoming revenue is recorded, this also makes it easier for businesses to seek financing opportunities from banks and investors.

Use with: Double-Entry Accounting: When performing accrual accounting, a double-entry accounting

system is highly recommended. Each transaction is classified as a debit and credit to different accounts

which makes it more appropriate than a single-entry accounting method.

iii. Modified Cash-Basis Accounting method

The modified cash-basis accounting method is a combination of cash and accrual accounting. All transactions and payments related to income and expenses are recorded once it occurs. Instead of choosing

one accounting method, this type lets businesses record short and long-term transactions and gets the best of

both worlds.

Cash-basis accounting lets businesses use a mix of accounts such as cash, liabilities, assets and accounts

payable.

Use with: Double-Entry Accounting: Much like accrual accounting, modified cash-basis accounting is used

with double-entry accounting.

2.2.1 Example of Cash Accounting

Under the cash accounting method, say Company A receives \$10,000 from the sale of 10 computers sold to

Company B on November 2, and records the sale as having occurred on November 2. The fact that Company

B in fact placed the order for the computers back on October 5 is deemed irrelevant, because it did not pay

for them until they were physically delivered on November 2. Under accrual accounting, by contrast,

Company A would have recorded the \$10,000 sale on October 5, even though no cash had yet changed

hands. Similarly, under cash accounting companies record expenses when they actually pay them, not when

they incur them. If Company C hires Company D for pest control on January 15, but does not pay the invoice

for the service completed until February 15, the expense would not be recognized until February 15 under

cash accounting. Under accrual accounting, however, the expense would be recorded in the books on January

15 when it was initiated.

Example of cash basis accounting entry for monthly income statement is given below:

2.3 Advantages of cash accounting and Limitations

2.3.1 Advantages: Cash accounting benefits are as below.

i. Simple to implement: Cash accounting is easy and no training is required to use the method from day 1 of

running your business. It isn't time-consuming and complicated either, so it is easy to get other people on

board to understand the process when compared to accrual accounting.

ii. Accurate cash flow insight: It shows the cash flow of the business at the moment. It doesn't factor in the past or the future but shows how much cash is available in a business at any given time. For start-ups and small businesses, this can be an excellent way to know how much cash they have at any time.

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iii. Simplifies tax returns: Cash accounting makes tax returns easy for some MSMEs. Let us say a client owes you money. As they have not paid yet, you have a lower tax burden for the current financial year because you won't account for it unless the cash enters your business. This is beneficial for large payments.

2.3.2: Limitations of Cash Accounting

i. A main drawback of cash accounting is that it may not provide an accurate picture of the liabilities that have been incurred (i.e. accrued) but not yet paid for, so that the business might appear to be better off than it really is. On the other hand, cash accounting also means that a business that has just completed a large job for which it is awaiting payment may appear to be less successful than it really is because it has expended the materials and labor for the job but not yet collected payment. Therefore, cash accounting can both overstate or understate the condition of the business if collections or payments happen to be particularly high or low in one period versus another.

ii. There are also some potentially negative tax consequences for businesses that adopt the cash accounting method. In general, businesses can only deduct expenses that are recognized within the current tax year.

iii. Inflow and outflow not properly captures. If a company incurs expenses in December 2019, but does not make payments against the expenses until January 2020, it would not be able to claim a deduction for the fiscal year ended 2019, which could significantly affect the business' bottom line. Likewise, a company that receives payment from a client in 2020 for services rendered in 2019 will only be allowed to include the revenue in its financial statements for 2020.

2.3.4 Cash Basis Accounting Vs Accrual Basis Accounting

Some of the differences between Cash and Accrual basis accounting

Cash Basis Accounting Accrual Basis Accounting

The simple system that keeps a

Complicated method.

record of business cash flow

Apt for small business, sole

Suitable for businesses that don't get paid right at
proprietorship firm that mostly
the moment.

deals with transactions in cash.

Gives a clear picture of the amount

Gives a clear picture of the correct financial
of cash in hand and the bank
position of a business;
account;

It doesn't reflect the money that is

It records money owed to you and the money you
owed to you or money you owe to
owe to others.

others.

2.3.5 Choosing the Best Accounting Method

Each accounting method comes with pros and cons. The method you choose will impact how
taxation is

reported and your overview of your business' financial situation.

If you're a large and complex business, accrual accounting will give you a better understanding of
your

business's revenue. For example, a marketing company may be paid periodically based on the
percentage of

work performed or compensation could wait until the entire project is completed. Financial
institutions and

investors may view this accounting method more favorably because it gives a better overview of its
revenue

and expenses in the long-term. Cash accounting method is ideal for small businesses which prefer a
straightforward way to measure income and expenses. However, revenue won't appear on the
ledger until the

payment is received. Nevertheless, businesses looking to combine the two methods can opt for
modified

cash-basis accounting. This allows them to view short-term cash transactions and long-term
financial

transactions related to their business.

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Note: When choosing an accounting method, be sure to educate yourself on its implications. The
ideal

method should provide you with an understanding of your business' financial health and
transactions. To

help make the process easier, businesses often choose to use accounting software.

2.3.6 Marginal Analysis

Marginal analysis is an examination of the associated costs and potential benefits of specific business

activities or financial decisions. The goal is to determine if the costs associated with the change in activity

will result in a benefit that is sufficient enough to offset them. Instead of focusing on business output as a

whole, the impact on the cost of producing an individual unit is most often observed as a point of comparison. Marginal analysis can also help in the decision-making process when two potential investments

exist, but there are only enough available funds for one. By analyzing the associated costs and estimated

benefits, it can be determined if one option will result in higher profits than another.

2.3.7 Marginal Analysis and Observed Change

From a microeconomic standpoint, marginal analysis can also relate to observing the effects of small changes

within the standard operating procedure or total outputs. For example, a business may attempt to increase

output by 1% and analyze the positive and negative effects that occur because of the change, such as changes

in overall product quality or how the change impacts the use of resources. If the results of the change are

positive, the business may choose to raise production by 1% again, and reexamine the results. These small

shifts and the associated changes can help a production facility determine an optimal production rate.

2.3.8 Marginal Cost vs. Marginal Benefit

Marginal cost is the cost to produce one additional unit of production. It is an important concept in cost

accounting as marginal cost helps determine the most efficient level of production for a manufacturing

process. It is calculated by determining what expenses are incurred if only one additional unit is

manufactured. A marginal benefit (or marginal product) is an incremental increase in a consumer's benefit in

using an additional unit of something. A marginal cost is an incremental increase in the expense a company

incurs to produce one additional unit of something. Marginal benefits normally decline as a consumer

decides to consume more and more of a single good. For example, a company is making fancy gadget that

are in high demand. Due to this demand, the company can afford machinery that reduces the average cost to

produce each gadget; the more they make, the cheaper they become. On average, it costs ₦50 to produce a

single gadget, but because of the new machinery, producing the 101 st gadget only costs ₦1. Therefore, the marginal cost of producing the 101st gadget is ₦1.

2.3.9 Marginal Cost Formula

Marginal cost is calculated as the total expenses required to manufacture one additional good. Therefore, it

can be measured by changes to what expenses are incurred for any given additional unit.

$$\text{Marginal Cost} = \text{Change in Total Expenses} / \text{Change in Quantity of Units Produced}$$

The formula above can be used when more than one additional unit is being manufactured. However,

management must be mindful that groups of production units may have materially varying levels of marginal cost.

2.3.10 Example of Marginal Analysis in Manufacturing

When a manufacturer wishes to expand its operations, either by adding new product lines or increasing the

volume of goods produced from the current product line, a marginal analysis of the costs and benefits is

necessary. Some of the costs to be examined include, but are not limited to, the cost of additional manufacturing equipment, any additional employees needed to support an increase in output, large facilities

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for manufacturing or storage of completed products, and as the cost of additional raw materials to produce the goods.

Once all of the costs are identified and estimated, these amounts are compared to the estimated increase in

sales attributed to the additional production. This analysis takes the estimated increase in income and

subtracts the estimated increase in costs. If the increase in income outweighs the increase in cost, the

expansion may be a wise investment.

For example, consider a hat manufacturer. Each hat produced requires seventy-five kobo (₦0.75) of plastic

and fabric. Your hat factory incurs ₦100 dollars of fixed costs per month. If you make 50 hats per month,

then each hat incurs ₦2 of fixed costs. In this simple example, the total cost per hat, including the plastic and

fabric, would be ₦2.75 ($\text{₦2.75} = \text{₦0.75} + (\text{₦100}/50)$). But, if you cranked up production volume and produced 100 hats per month, then each hat would incur ₦1 dollar of fixed costs because fixed costs are

spread out across units of output. The total cost per hat would then drop to ₦1.75 ($\text{₦1.75} = \text{₦0.75} +$

($\$100/100$)). In this situation, increasing production volume causes marginal costs to go down.

2.3.11 Why Is Marginal Analysis Important

Marginal analysis is important because it identifies the most efficient use of resources. An activity should

only be performed until the marginal revenue equals the marginal cost; beyond this point, it will cost more

for every unit that the benefit received for every unit.

2.4 Cash Flows and Cash Flow Diagrams

2.4.1 Cash flow

Cash flows are a fundamental tool in engineering economic analysis. Cash flows represent transactions in

which money changes hands between two or more parties for example lending \$10.00 to a friend or making a

payment on a car loan, for example. Representing transactions as cash flows makes it easier to keep track of

the important information included in the transactions. Specifically, there are two characteristics of financial

transactions that are indicated in cash flows:

i) Value – the magnitude of the transaction being described. This is dependent on two factors: the amount of

money or currency changing hands (e.g, a dollar value) and the direction in which the money is flowing (the

orientation of the cash flow). We represent financial gains (also called receipts or income) as positive in

value, and financial losses (also called disbursements or expenses) as negative in value.

ii) Timing – the time or period in which the cash flow occurs. Often, periods are set to coincide with interest

periods, which will be discussed further in the Time Value of Money chapter. Typically, periods are in

increments of months, quarters (1/4 of a year), semi-annual, or annual but other time increments may also be

used. Let's use the example of someone lending \$10.00 to a friend. We'll assume Riley lends his friend Chris

\$10.00 on January 1, and Chris pays back the \$10.00 on February 1. There are two transactions in this

example: the initial lending on January 1, and the repayment on February 1. Each can be represented by its

own cash flow. From Riley's perspective, as the lender, the two cash flows would look like so:

Table 2: Cash flows from Riley's Perspective

Cash Flow Timing Value

Lend \$10 to Chris January 1 -\$10.00

Receive \$10 from Chris February 1 +\$10.00

The table in Table 2 gives a concise summary of the two cash flows involved in the example, from the lender's perspective. The approach would change if we made the same table from Chris's (the borrower's) perspective

Table 3: Cash flows from Chris' Perspective

Cash Flow Timing Value

Receive \$10 from Riley January 1 +\$10.00

Repay \$10 to Riley February 1 -\$10.00

As we can see by comparing Tables 2 and 3, changing perspectives changes the signage of the cash flows. This is because when one party gives money to the other, the recipient's total assets increase, and the donor's assets decrease.

The previous example shows how cash flows can be used to summarize the important information in financial transactions. When conducting an analysis in a spreadsheet it is common to list cash flows in tabular format. When trying to visualize or explain the financial transactions in a particular analysis, cash flows can be represented in a much simpler way known as the cash flow diagram.

2.4.2 Cash Flow Diagram

Cash Flow Diagrams are simple graphical representations of financial transactions. The diagrams consist of arrows, such as in the diagram shown below.

Figure 1 Cash Flow Diagram

Cash flows like the ones shown in the above figure are typical. There are some basic rules for creating cash flow diagrams:

- i) Time is represented by a horizontal line marked with the number of periods in the analysis. The choice of time interval will reflect the project or transactions being considered.
- ii) The horizontal position of each arrow indicates the timing of that cash flow.
- iii) Upward arrows represent positive cash flows, also known as inflows, income, or receipts.
- iv) Downward arrows represent negative cash flows, also known as outflows, disbursements, or expenses.
- v) Each arrow represents the net cash flow in that period (receipts - disbursements). There is only one cash flow arrow for each period representing this net value.

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Let's return to the previous example: Riley lending \$10.00 to Chris. We can draw out the cash flows

graphically as a cash flow diagram. Just like with the tabular form of the cash flows, we can represent the

cash flows in two different diagrams: one from Riley's perspective and one from Chris's perspective. If we

set January to be the first period, and February to be the second period, the two diagrams would look like so:

Figure 2: Cash flow diagrams of a lending agreement

These diagrams are really just graphical representations of financial transactions. As one can see, the

diagrams for Riley and Chris give the same information as the tables we showed before, but it may be less

time-consuming to draw these simple diagrams than it is to write out an entire table. The movement of

money is also easier to show graphically. Note that only one of the diagrams is required to represent the

problem. You would simply choose to display the project from either Riley's perspective or from Chris's

perspective.

Note that this difference in perspective does not affect the equivalence of cash flows. The different perspectives affect the signs of the cash flows (i.e. positive or negative), but the net effect remains the same.

In this example, both parties ended up with no net loss or gain after Riley was repaid, because no interest was

charged on the loan. If Riley had charged Chris \$1.00 in interest, then Riley would have earned a net profit of

\$1.00 and Chris would have lost \$1.00 in total. The money is all accounted for, regardless of who the

observer is.

2.4.3 End-of-Period Convention

In practice, cash flows can occur at any time within a period. However, for simplicity, we commonly assume

the end-of-period convention – the assumption that all cash flows occurring within a period are moved to

the end of the period. The following figure graphically demonstrates the end-of-period convention.

Fig. 4: End of Period Diagram

In the figure 4, there are several cash flows at different times within the same period. Using the end-of-

period convention, we sum these cash flows together and move them to the end of the interest period as one

net cash flow. While this assumption could introduce some discrepancies between the model and real-world

results, it simplifies calculations greatly.

2.5 Time value of money

The time value of money (TVM) is the concept that a sum of money is worth more now than the same sum

will be at a future date due to its earnings potential in the interim. The time value of money is a core principle

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of finance. A sum of money in the hand has greater value than the same sum to be paid in the future. The

time value of money is also referred to as the present discounted value. The time value of money means that

a sum of money is worth more now than the same sum of money in the future. The principle of the time

value of money means that it can grow only through investing so a delayed investment is a lost opportunity.

2.5.1 Understanding the Time Value of Money (TVM)

Investors prefer to receive money today rather than the same amount of money in the future because a sum of

money, once invested, grows over time. For example, money deposited into a savings account earns interest.

Over time, the interest is added to the principal, earning more interest. That's the power of compounding

interest. If it is not invested, the value of the money erodes over time. If you hide \$1,000 in a mattress for

three years, you will lose the additional money it could have earned over that time if invested. It will have

even less buying power when you retrieve it because inflation reduces its value.

As another example, say you have the option of receiving \$10,000 now or \$10,000 two years from now.

Despite the equal face value, \$10,000 today has more value and utility than it will two years from now due

to the opportunity costs associated with the delay. In other words, a delayed payment is a missed opportunity.

The time value of money has a negative relationship with inflation. Remember that inflation is an increase

in the prices of goods and services. As such, the value of a single dollar goes down when prices rise, which

means you can't purchase as much as you were able to in the past. Note the followings:

2.5.2 Time Value of Money Formula

The most fundamental formula for the time value of money takes into account the following: the future

value of money, the present value of money, the interest rate, the number of compounding periods per year,

and the number of years.

Based on these variables, the formula for TVM is:

□

$$FV = PV \times (1 + i)^n$$

Where:

FV=Future value of money

PV=Present value of money

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i=Interest rate

n=Number of compounding periods per year

t=Number of years

Keep in mind, though that the TVM formula may change slightly depending on the situation. For example,

in the case of annuity or perpetuity payments, the generalized formula has additional or fewer factors.

The time value of money doesn't take into account any capital losses that you may incur or any negative

interest rates that may apply. In these cases, you may be able to use negative growth rates to calculate the

time value of money

2.5.3 Examples of Time Value of Money

Here's a hypothetical example to show how the time value of money works. Let's assume a sum of \$10,000

is invested for one year at 10% interest compounded annually. The future value of that money is: 10%

$$FV = \$10,000 \times (1 + 0.10)^1 = \$11,000$$

The formula can also be rearranged to find the value of the future sum in present-day dollars. For example,

the present-day dollar amount compounded annually at 7% interest that would be worth \$5,000 one year

from today is:

$$PV = \frac{FV}{(1 + i)^n}$$

$$PV = \frac{\$5,000}{(1 + 0.07)^1}$$

$$PV = [\$5,000 / 1.07] = \$4,673$$

$$1 + 0.07$$

Effect of Compounding Periods on Future Value

The number of compounding periods has a dramatic effect on the TVM calculations. Taking the \$10,000

example above, if the number of compounding periods is increased to quarterly, monthly, or daily, the

ending future value calculations are:

$$10\% \text{ } 4 \times 1$$

$$(a) \text{ Quarterly Compounding: } FV = \$10,000 \times (1 + 0.10/4)^4 = \$11,038$$

4

10% 12×1

(b) Monthly Compounding: $100 = \$10,000 \times (1 +) = 11,047$

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10%

(c) Daily Compounding: $FV = \$10,000 (1 + 365)^{365 \times 1} = \$11,052$

This shows that the TVM depends not only on the interest rate and time horizon but also on how many

times the compounding calculations are computed each year.

2.5.4 How Does the Time Value of Money Relate to Opportunity Cost

The opportunity cost of a choice is the value of the best alternative forgone where, given limited resources,

a choice needs to be made between several mutually exclusive alternatives. Hence, opportunity cost is the

value of what you lose when choosing between two or more options. Opportunity cost is key to the concept

of the time value of money. Money can grow only if it is invested over time and earns a positive return.

Money that is not invested loses value over time. Therefore, a sum of money that is expected to be paid in

the future, no matter how confidently it is expected, is losing value in the meantime.

2.5.5 Why Is the Time Value of Money Important

The concept of the time value of money can help guide investment decisions. For instance, suppose an

investor can choose between two projects: Project A and Project B. They are identical except that Project A

promises a \$1 million cash payout in year one, whereas Project B offers a \$1 million cash payout in year

five. The payouts are not equal. The \$1 million payout received after one year has a higher present value than the \$1 million payout after five years.

It would be hard to find a single area of finance where the time value of money does not influence the

decision-making process. The time value of money is the central concept in discounted cash flow (DCF)

analysis, which is one of the most popular and influential methods for valuing investment opportunities. It is

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also an integral part of financial planning and risk management activities. Pension fund managers, for

instance, consider the time value of money to ensure that their account holders will receive adequate funds

in retirement.

2.5.6 What Impact Does Inflation Have on the Time Value of Money

The value of money changes over time and there are several factors that can affect it. Inflation, which is the general rise in prices of goods and services, has a negative impact on the future value of money. That's because when prices rise, your money only goes so far. Even a slight increase in prices means that your purchasing power drops. So that dollar you earned in 2015 and kept in your bank buys less today than it would have back then.

2.5.7 How Do You Calculate the Time Value of Money

The time value of money takes several things into account when calculating the future value of money, including the present value of money (PV), the number of compounding periods per year (n), the total number of years (t), and the interest rate (i). You can use the following formula to calculate the time value of money:

$$FV = PV \times [1 + (i / n)]^{(n \times t)}$$

The Bottom Line: The future value of money isn't the same as present-day dollars. And the same is true about money from the past. This phenomenon is known as the time value of money. Businesses can use it to gauge the potential for future projects. And as an investor, you can use it to pinpoint investment opportunities. Put simply, knowing what TVM is and how to calculate it can help you make sound decisions about how you spend, save, and invest money.

2.5.8 Simple Interest

Simple Interest (S.I.) is the method of calculating the interest amount for a particular principal amount of money at some rate of interest. Interest is the cost of borrowing money. Typically expressed as a percentage, it amounts to a fee or charge that the borrower pays the lender for the financed sum.

Its formula: $SI = (PRT) / 100$

$$A = P (1 + RT)$$

A = final amount

P = initial principal balance

R = annual interest rate

T = time (in years)

For example, when a person takes a loan of #5000, at a rate of 10 p.a. for two years, the person's interest for two years will be S.I. on the borrowed money.

2.5.9 Compound interest

Compound interest is the interest calculated on the principal and the interest accumulated over the previous

period. The compound interest is the interest-based on the initial principal amount and the interest collected over

the period of time. The compound interest formula is given below:

Compound Interest = Amount – Principal

Here, the amount is given by;

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Where,

- A = amount
- P = principal
- r = rate of interest
- n = number of times interest is compounded per year
- t = time (in years)

Alternatively, we can write the formula as given below:

$CI = A - P$

And

This formula is also called periodic compounding formula.

- A represents the new principal sum or the total amount of money after compounding period
- P represents the original amount or initial amount
- r is the annual interest rate
- n represents the compounding frequency or the number of times interest is compounded in a year
- t represents the number of years

It is to be noted that the above formula is the general formula for the number of times the principal is

compounded in a year. If the interest is compounded annually, the amount is given as:

Thus, the compound interest rate formula can be expressed for different scenarios such as the interest rate is

compounded yearly, half-yearly, quarterly, monthly, daily, etc.

NOTE: From the data, it is clear that the interest rate for the first year in compound interest is the same as

that in simple interest. $PRT/100$. Other than the first year, the interest compounded annually is always greater

than that in simple interest.

2.5.9 Formula for Periodic Compounding Rate

The total accumulated value, including the principal P plus compounded interest I, is given by the formula:

$$P' = P[1 + (r/n)]^{nt}$$

Here,

P = Principal

P' = New principal

r = Nominal annual interest rate

n = Number of times the interest is compounding

t = Time (in years)

In this case, compound interest is:

$$CI = P' - P$$

$$CI = P[1 + (r/n)]^{nt} - P$$

How to Calculate Compound Interest

Let us understand the process of calculating compound interest with the help of the below example.

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Example: What amount is to be repaid on a loan of Rs. 12000 for one and half years at 10% per annum

compounded half yearly?

Solution:

For the given situation, we can calculate the compound interest and total amount to be repaid on a loan in

two ways. In the first method, we can directly substitute the values in the formula. In the second method,

compound interest can be obtained by splitting the given time bound into equal periods.

This can be well understood with the help of the table given below.

Compound Interest Solved Examples

As mentioned above, compound interest has many applications in real-life. Let us solve various examples

based on these applications to understand the concept in a better manner.

Increase or Decrease in Population

Examples 1: A town had 10,000 residents in year 2000. Its population declines at a rate of 10% per annum.

What will be its total population in 2005?

Solution: The population of the town decreases by 10% every year. Thus, it has a new population every year.

So the population for the next year is calculated on the current year population.

For the decrease, we have the formula $A = P(1 - R/100)^n$

Therefore, the population at the end of 5 years = $10000(1 - 10/100)^5$

$$= 10000(1 - 0.1)^5 = 10000 \times 0.95 = 5904 \text{ (Approx.)}$$

Examples 2: The count of a certain breed of bacteria was found to increase at the rate of 2% per hour. Find

the bacteria at the end of 2 hours if the count was initially 600000.

Solution: Since the population of bacteria increases at the rate of 2% per hour, we use the formula

$$A = P(1 + R/100)^n$$

Thus, the population at the end of 2 hours = $600000(1 + 2/100)^2$

$$= 600000(1 + 0.02)^2 = 600000(1.02)^2 = 624240$$

Examples 3: The price of a radio is Rs. 1400 and it depreciates by 8% per month. Find its value after 3 months.

Solution: For the depreciation, we have the formula $A = P(1 - R/100)^n$.

Thus, the price of the radio after 3 months = $1400(1 - 8/100)^3$

= $1400(1 - 0.08)^3 = 1400(0.92)^3 = \text{Rs. } 1090 \text{ (Approx.)}$

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2.5.10 Compound Interest and Simple Interest

Now, let us understand the difference between the amount earned through compound interest and simple

interest on a certain amount of money, say Rs. 100 in 3 years and the rate of interest is 10% p.a.

Below table shows the process of calculating interest and total amount.

2.5.11 Compound Interest Problems

Try solving the below questions on compound interest.

1. A sum of Rs.10000 is borrowed by Akshit for 2 years at an interest of 10% compounded annually. Find

the compound interest and amount he has to pay at the end of 2 years.

2. What is the compound interest (CI) on Rs.5000 for 2 years at 10% per annum compounded annually?

3. What is the compound interest to be paid on a loan of Rs.2000 for $3/2$ years at 10% per annum compounded half-yearly?

4. What is the least number of complete years in which a sum of money put out at 20% compound interest

will be more than doubled?

5. Heera invests Rs. 20,000 at the beginning of every year in a bank and earns 10 % annual interest,

compounded at the end of the year. What will be her balance in the bank at the end of three years?

6. What is the difference between the compound interests on Rs. 5000 for one and half years at 4% per

annum compounded yearly and half-yearly?

2.5.12 Questions and Answers on Compound Interest

Q1.What is Compound interest?

Ans. Compound interest is the interest calculated on the principal and the interest accumulated over the

previous period.

Q2, How do you calculate compound interest?

Ans. Compound interest is calculated by multiplying the initial principal amount (P) by one plus the annual

interest rate (R) raised to the number of compound periods (nt) minus one. That means, $CI = P[(1 + R)^{nt} - 1]$

Here,

P = Initial amount

R = Annual rate of interest as a percentage

n = Number of compounding periods in a given time

Q3 Who benefits from compound interest?

Ans. The investors benefit from the compound interest since the interest pair here on the principle plus on the interest which they already earned.

Q4. What is interest compounded quarterly formula?

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The formula for interest compounded quarterly is given by:

$$A = P(1 + (R/4)/100)^{4T}$$

Q5. How do you find the compound interest rate?

Ans. The compound interest rate can be found using the formula,

$$A = P(1 + r/n)^{nt}$$

A = Total amount

P = Principal

r = Annual nominal interest rate as a decimal

n = Number of compounding periods

t = Time (in years)

Thus, compound interest (CI) = A - P

Q6. What is the formula of compound interest with an example?

Ans. The compound interest formula is given below:

Compound Interest = Amount - Principal

Where the amount is given by:

$$A = P(1 + r/n)^{nt}$$

P = Principal

r = Annual nominal interest rate as a decimal

n = Number of compounding periods

t = Time (in years)

For example, If Mohan deposits Rs. 4000 into an account paying 6% annual interest compounded quarterly,

and then the money will be in his account after five years can be calculated as:

Substituting, P = 4000, r = 0.06, n = 4, and t = 5 in $A = P(1 + r/n)^{nt}$, we get A = Rs. 5387.42

Q7. What is the compounded daily formula?

Ans. The compound interest formula when the interest is compounded daily is given by:

$$A = P(1 + r/365)^{365 * t}$$

2.5.13 Difference between Simple and Compound Interest

Simple Interest: Simple interest can be defined as the principal amount of a loan or deposit a person makes

into their bank account.

Compound Interest: Compound interest is the interest that accumulates and compounds over the principal

amount.

Below you can find the key differences between Simple Interest and Compound Interest in the tabular

column below:

Simple Interest and Compound Interest Differences

Parameter Simple Interest Compound Interest

Definition Simple Interest can be defined as Compound Interest can be defined as when the the sum paid back for using the sum principal amount exceeds the due date for borrowed money over a fixed payment, along with the rate of interest for a period of time. period of time.

Formula S.I. = $(P \times T \times R) / 100$ C.I. = $P(1+R/100)^t - P$

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Simple Interest and Compound Interest Differences

Return The return is much lesser when The return is much higher.

Amount compared to compound interest.

Principal The principal amount is constant. The principal amount keeps on varying during Amount the entire borrowing period.

Growth The growth remains quite uniform The growth increases quite rapidly in this in this method. method.

Interest The interest charged on is for the The interest charged on it is for the principal Charged principal amount. and accumulated interest.

Solved Examples

Q.1: Olumide borrowed ₦50,000 for 3 years at a rate of 3.5% per annum. Find the simple interest.

Solution: Given,

$P = ₦ 50,000$

$R = 3.5\%$

$T = 3 \text{ years}$

$SI = (P \times R \times T) / 100$

$SI = (50,000 \times 3.5 \times 3) / 100 = 5250$

Q.2: The count of a population of men was found to increase at the rate of 2% per hour. Find the count at the end of 2 hours if the initial count was 600000.

Solution: Since the population of men increases at the rate of 2% per hour, we use the formula

$A = P(1 + R/100)^n$

Thus, the population at the end of 2 hours = $600000(1 + 2/100)^2$

$= 600000(1 + 0.02)^2$

$= 600000(1.02)^2$

$= 624240$

Q3 What is the main difference between simple interest and compound interest?

Simple interest is computed on the principal amount or loan amount whereas compound interest is computed

based on the principal amount as well as the interest accumulated for a certain period or previous period.

Q4 What is the formula for simple interest?

The formula for simple interest is given by:

$$SI = (P \times R \times T)/100$$

where SI = Simple Interest

P = Principal Amount

R = Rate of interest

T = Time duration in years

Q5 What is the formula for compound interest?

The formula for compound interest is given by:

$$CI = \text{Amount} - \text{Principal}$$

$$\text{and Amount} = P(1+r/n)^{nt}$$

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Q6 What is the formula for the amount if it is compounded annually?

If the amount is compounded annually, the amount is given as:

$$A = P(1+R/100)^t$$

2.6 Discounted Cash Flow (DCF)

Discounted cash flow (DCF) is a technique that determines the present value of future cash flows. This

approach can be used to derive the value of an investment. Under the DCF method, one applies a discount

rate to each periodic cash flow that is derived from an entity's cost of capital. Multiplying this discount by

each future cash flow results in an amount that is, in aggregate, the present value of all future cash flows.

By calculating the discounted cash flows for a number of different investment choices, one can select the

alternative that results in the greatest discounted cash flows. This concept is useful for calculating the value

of a prospective acquisition, of a possible annuity investment, or of a fixed asset purchase.

The foundation of discounted cash flow analysis is the concept that cash received today is more valuable than

cash received at some point in the future. The reason is that someone who agrees to receive payment at a

later date foregoes the ability to invest that cash right now. The only way for someone to agree to a delayed

payment is to pay them for the privilege, which is known as interest income.

The purpose of DCF analysis is to estimate the money an investor would receive from an investment,

adjusted for the time value of money.

The time value of money assumes that a dollar that you have today is worth more than a dollar that you

receive tomorrow because it can be invested. As such, a DCF analysis is useful in any situation where a

person is paying money in the present with expectations of receiving more money in the future.

For example, assuming a 5% annual interest rate, \$1 in a savings account will be worth \$1.05 in a year.

Similarly, if a \$1 payment is delayed for a year, its present value is 95 cents because you cannot transfer it

to your savings account to earn interest. Discounted cash flow analysis finds the present value of expected

future cash flows using a discount rate. Investors can use the concept of the present value of money to

determine whether the future cash flows of an investment or project are greater than the value of the initial

investment. If the DCF value calculated is higher than the current cost of the investment, the opportunity

should be considered. If the calculated value is lower than the cost, then it may not be a good opportunity,

or more research and analysis may be needed before moving forward with it.

To conduct a DCF analysis, an investor must make estimates about future cash flows and the ending value

of the investment, equipment, or other assets. The investor must also determine an appropriate discount rate

for the DCF model, which will vary depending on the project or investment under consideration. Factors

such as the company or investor's risk profile and the conditions of the capital markets can affect the

discount rate chosen. If the investor cannot estimate future cash flows, or the project is very complex, DCF

will not have much value and alternative models should be employed.

2.6.1 Discounted Cash Flow Formula

The formula for DCF is:

$$CF_1 + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \dots + \frac{CF_n}{(1+r)^n}$$

$$DCF = \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \dots + \frac{CF_n}{(1+r)^n}$$

$$DCF = \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \dots + \frac{CF_n}{(1+r)^n}$$

where:

CF1=The cash flow for year one

CF2=The cash flow for year two

CF3=The cash flow for year three

CFn=The cash flow for additional years

r=The discount rate

2.6.1 Advantages and Disadvantages of DCF

i. Advantages

- a. Discounted cash flow analysis can provide investors and companies with an idea of whether a proposed investment is worthwhile
- b. It is analysis that can be applied to a variety of investments and capital projects where future cash flows can be reasonably estimated.
- c. Its projections can be tweaked to provide different results for various what if scenarios. This can help users account for different projections that might be possible.

ii. Disadvantages

- a. The major limitation of discounted cash flow analysis is that it involves estimates, not actual figures.

So the result of DCF is also an estimate. That means that for DCF to be useful, individual investors and companies must estimate a discount rate and cash flows correctly.

- b. Furthermore, future cash flows rely on a variety of factors, such as market demand, the status of the economy, technology, competition, and unforeseen threats or opportunities. These can't be quantified exactly. Investors must understand this inherent drawback for their decision-making.
 - c. DCF shouldn't necessarily be relied on exclusively even if solid estimates can be made.
- Companies

and investors should consider other, known factors as well when sizing up an investment opportunity. In addition, comparable company analysis and precedent transactions are two other, common valuation methods that might be used.

2.6.3 Example of DCF

When a company analyzes whether it should invest in a certain project or purchase new equipment, it

usually uses its weighted average cost of capital (WACC) as the discount rate to evaluate the DCF. The WACC incorporates the average rate of return that shareholders in the firm are expecting for the given year.

For example, say that your company wants to launch a project. The company's WACC is 5%. That means

that you will use 5% as your discount rate. The initial investment is \$11 million, and the project will last for

five years, with the following estimated cash flows per year.

Cash Flow

Year Cash Flow

1 \$1 million

2 \$1 million

3 \$4 million

4 \$4 million

5 \$6 million

Using the DCF formula, the calculated discounted cash flows for the project are as follows.

Discounted Cash Flow

Year Cash Flow Discounted Cash Flow (nearest \$)

1 \$1 million \$952,381

2 \$1 million \$907,029

3 \$4 million \$3,455,350

4 \$4 million \$3,290,810

5 \$6 million \$4,701,157

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Adding up all of the discounted cash flows results in a value of \$13,306,727. By subtracting the initial

investment of \$11 million from that value, we get a net present value (NPV) of \$2,306,727. The positive

number of \$2,306,727 indicates that the project could generate a return higher than the initial cost—a

positive return on the investment. Therefore, the project may be worth making. If the project had cost \$14

million, the NPV would have been -\$693,272. That would indicate that the project cost would be more than

the projected return. Thus, it might not be worth making.

If a person owns \$10,000 now and invests it at an interest rate of 10%, then she will have earned \$1,000 by

having use of the money for one year. If she were instead to not have access to that cash for one year, then

she would lose the \$1,000 of interest income. The interest income in this example represents the time value

of money. Two analysis methods that employ the discounted cash flow concept are net present value and the

internal rate of return.

2.7 Net Present Value

Net present value (NPV) analysis is useful for determining the current value of a stream of cash flows that

extend out into the future. It can also be used to compare several such cash flows to decide which has the

largest present value. NPV is commonly used in the analysis of capital purchasing requests, to see if an

initial payment for fixed assets and other expenditures will generate net positive cash flows.

2.7.1 NPV formula

The NPV formula is a method of determining the profitability of an investment by discounting the future

cash flows of the investment to today's value. Unlike the internal rate of return (IRR), the NPV calculation

formula requires a discount rate. It also depends on the investment's intervals and number of future cash

flows. This means the formula for calculating the NPV for a short-term project with one cash flow differs

from that of a multiyear investment with multiple cash flows. There are two NPV formulas typically used,

including:

1. NPV formula for an investment with a single cash flow:

When calculating the NPV for a short-term project with a single cash flow, the only variables required to

get the present value are the cash flow, the cash flow period and the discount rate. Here's the NPV formula

for a one-year project with a single cash flow:

$$NPV = [\text{cash flow} / (1+i)^t] - \text{initial investment}$$

Where i = discount rate, and t = the number of time periods.

2. NPV formula for a project with multiple cash flows and a longer duration

The formula for longer-term investments with multiple cash flows is almost the same, except you discount

each cash flow individually and then add them together. Here's the NPV formula for a longer-term project

with multiple cash flows:

$$NPV = \text{sum of the present value of expected cash flows} - \text{initial investment}$$

Annual net cash flows: You can estimate each year's net cash flows by adding the expected cash inflows

from projected revenues to potential savings in labor, materials and other components of the initial project

cost. Then, deduct the costs incurred over the new project, or the cash outflows during a single period. You

can also add the tax savings on the depreciation of project assets to the expected cash inflows.

Cash inflows

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must be positive, while cash outflows must be negative. If your expected cash inflows exceed outflows, it

results in a net cash inflow. You can expect net cash outflow if the expected cash outflow exceeds the

expected inflows.

Interest rate: The interest rate is vital to the calculation of NPV. Most managers use the discount rate to

represent the interest rate, also known as the cost of capital, cutoff rate, the required rate of return or hurdle

rate. The interest rate or discount rate is the cost of capital or return that you could earn in an alternative

investment. You derive the interest rate by comparing the rate of return of alternative investments or projects with similar initial costs. The NPV of projects with a constant payment size and a fixed interest rate is usually straightforward and easier to calculate.

Time period: The time period is the interval at which you invest new cash flows into the new project. You determine the period used in the calculation, which can be daily, monthly or whichever time frame you wish to measure. Businesses may choose to use an annual period to simplify transactions.

2.7.2 How to calculate NPV

To estimate the net present value of a proposed investment, determine the expected net present value of the future cash flows from the investment and deduct the project's initial investment. You can accept the project if the NPV result is zero or positive. A negative NPV result means the project won't be profitable and investors may reject it.

Here are the steps for calculating NPV:

- i. Find today's value of the expected cash flow
- ii. To calculate the present value, gather the following information:
- iii. Period: This is the period for which you require the calculation. A typical example is 12 months.
- iv. Rate per period: This is the interest rate of the period.
- v. Compounding: You may choose to add compounding, but if not, use one.
- vi. Cash flow: This includes the cash flow at the beginning and end of the chosen period.

Below are examples of how to use the NPV formula to calculate the NPV of capital projects:

Single cash flow NPV example

Dexable Inc. is planning a project with an initial investment of \$5,000. The company projects the investment to generate a cash flow of \$5,600 in the next year. The NPV assumes that the required rate of return is 10% and there's no salvage value at the end of the project. To calculate whether it can expect a profit, Dexable Inc. uses the following formula:

$$\text{NPV} = [\$5,600 / (1+0.1)^1] - \$5,000 = \$90.90$$

This NPV shows that the project is profitable, so the managers can accept it.

Multiple cash flow NPV example

Project R requires an initial investment of \$45,000 and expects to generate \$30,000 annually for two years, with an 8% discount rate. To estimate the NPV, the company that wants to invest in Project R finds the individual NPV of both cash flows and adds the results. Then, it deducts the initial investment from the sum

of the NPVs. Here are the company's calculations for the NPV, which shows a profit:

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$$\text{NPV of Project R} = [\$30,000 / (1+0.08)^1] + [\$30,000 / (1+0.08)^2] - \$45,000$$

$$\text{NPV} = (\$27,777.78 + \$25,720.17) - \$45,000 = \$8,497.95$$

The results show the NPV of Project R is positive and the investment should turn a profit.

NPV loss prediction example

The initial investment for Project Nathan is \$50,000. It expects to provide revenues of \$13,000, \$26,000 and

\$23,000 for the first, second and third years, respectively. The cost of capital is 13%.

The following calculations show that these revenues result in a loss:

$$\text{NPV of Project Nathan} = [\$13,000 / (1+0.13)^1] + [\$26,000 / (1+0.13)^2] + [\$23,000 / (1+0.13)] - \$50,000$$

$$\text{NPV} = (\$11,504.43 + \$18,012.37 + \$15,940.15) - \$50,000 = \$45,456.95 - \$50,000 = -\$4,543.05$$

The NPV for Project Nathan shows that the investment results in a loss, so the planners should reject the project.

2.8 Internal Rate of Return

The internal rate of return (IRR) is the rate of return at which the present value of a series of future cash

flows equals the present value of all associated costs. IRR is commonly used in capital budgeting to discern

the rate of return on the estimated cash flows arising from an expected investment.

The internal rate of return (IRR) is a metric used in financial analysis to estimate the profitability of potential investments. IRR is a discount rate that makes the net present value (NPV) of all cash flows equal

to zero in a discounted cash flow analysis. IRR calculations rely on the same formula as NPV does.

Keep in

mind that IRR is not the actual dollar value of the project. It is the annual return that makes the NPV equal

to zero.

Generally speaking, the higher an internal rate of return, the more desirable an investment is to undertake.

IRR is uniform for investments of varying types and, as such, can be used to rank multiple prospective

investments or projects on a relatively even basis. In general, when comparing investment options with

other similar characteristics, the investment with the highest IRR probably would be considered the best.

Note the followings:

- The internal rate of return (IRR) is the annual rate of growth that an investment is expected to generate.

- IRR is calculated using the same concept as net present value (NPV), except it sets the NPV equal to zero.
- The ultimate goal of IRR is to identify the rate of discount, which makes the present value of the sum of annual nominal cash inflows equal to the initial net cash outlay for the investment.
- IRR is ideal for analyzing capital budgeting projects to understand and compare potential rates of annual return over time.
- In addition to being used by companies to determine which capital projects to use, IRR can help investors determine the investment return of various assets. The project having the highest IRR is selected for investment purposes.

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Formula and Calculation for IRR

The formula and calculation used to determine this figure are as follows:

□□

$$0 = NPV = \sum_{t=0}^n \frac{C_t}{(1+IRR)^t} - C_0$$

$$(1+IRR)^t$$

where: □□ = Net cash inflow during the period t

□0 = Total initial investment costs

IRR = The internal rate of return

t = The number of time periods

How to Calculate IRR

1. Using the formula, one would set NPV equal to zero and solve for the discount rate, which is the IRR.
2. The initial investment is always negative because it represents an outflow.
3. Each subsequent cash flow could be positive or negative, depending on the estimates of what the project delivers or requires as a capital injection in the future.
4. However, because of the nature of the formula, IRR cannot be easily calculated analytically and instead must be calculated iteratively through trial and error or by using software programmed to calculate IRR (e.g., using Excel)

IRR Example

Assume a company is reviewing two projects. Management must decide whether to move forward with one,

both, or neither. Its cost of capital is 10%. The cash flow patterns for each are as follows:

Project A

- Initial Outlay = \$5,000
- Year one = \$1,700
- Year two = \$1,900
- Year three = \$1,600
- Year four = \$1,500

- Year five = \$700

Project B

- Initial Outlay = \$2,000
- Year one = \$400
- Year two = \$700
- Year three = \$500
- Year four = \$400
- Year five = \$300

The company must calculate the IRR for each project. Initial outlay (period = 0) will be negative.
Solving

for IRR is an iterative process using the following equation:

$$\$0 = \sum C F_t \div (1 + IRR)^t$$

where:

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- CF = net cash flow
- IRR = internal rate of return
- t = period (from 0 to last period)

Hence,

$$\$0 = [\text{initial outlay} \times (-1)] + CF_1 \div (1 + IRR)^1 + CF_2 \div (1 + IRR)^2 + \dots + CF_X \div (1 + IRR)^X$$

Using the above examples, the company can calculate IRR for each project as:

IRR Project A

$$\begin{aligned} \$0 = & (-\$5,000) + \$1,700 \div (1 + IRR)^1 + \$1,900 \div (1 + IRR)^2 + \$1,600 \div (1 + IRR)^3 + \$1,500 \div (1 + IRR)^4 \\ & + \$700 \div (1 + IRR)^5 \end{aligned}$$

$$IRR \text{ Project A} = 16.61 \%$$

IRR Project B

$$\begin{aligned} \$0 = & (-\$2,000) + \$400 \div (1 + IRR)^1 + \$700 \div (1 + IRR)^2 + \$500 \div (1 + IRR)^3 + \$400 \div (1 + IRR)^4 \\ & + \$300 \div (1 + IRR)^5 \end{aligned}$$

$$IRR \text{ Project B} = 5.23 \%$$

Given that the company's cost of capital is 10%, management should proceed with Project A and reject

Project B.

2.8.1 Payback Period

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2.9 Taxation

A tax is a compulsory financial charge or some other type of levy imposed on a taxpayer (an individual or

legal entity) by a governmental organization in order to collectively fund government spending, public

expenditures, or as a way to regulate and reduce negative externalities. Tax compliance refers to policy

actions and individual behaviour aimed at ensuring that taxpayers are paying the right amount of tax at the

right time and securing the correct tax allowances and tax reliefs.

Taxation is a term for when a taxing authority, usually a government, levies or imposes a financial obligation

on its citizens or residents. Hence, Taxation occurs when a government or other authority requires that a fee

be paid by citizens and corporations, to that authority. The fee is involuntary, and as opposed to other

payments, not linked to any specific services that has been or will be provided. Taxation is differentiated

from other forms of payment, such as market exchanges, in that taxation does not require consent and is not

directly tied to any services rendered. The government compels taxation through an implicit or explicit threat

of force. Taxation is legally different than extortion because the imposing institution is a government, not

private actors.

2.9.1 Purposes and Justifications for Taxation

The most basic function of taxation is to fund government expenditures. Varying justifications and explanations for taxes have been offered throughout history. Early taxes were used to support the ruling

classes, build defenses etc. Later justifications have been offered across utilitarian, economic, or moral

considerations. Proponents of progressive levels of taxation on high-income earners argue that taxes

encourage a more equitable society. Higher taxes on specific products and services, such as tobacco or

gasoline, have been justified as a deterrent to consumption.

2.9.2 Different Types of Taxation

As mentioned above, taxation applies to all different types of levies. These can include but are not limited to:

i. Income tax: Governments impose income taxes on financial income generated by all entities within their

jurisdiction, including individuals and businesses.

ii. Corporate tax: This type of tax is imposed on the profit of a business.

Capital gains: A tax on capital gains is imposed on any capital gains or profits made by people or businesses

from the sale of certain assets including stocks, bonds, or real estate.

iii. Property tax: A property tax is assessed by a local government and paid for by the owner of a property. This

tax is calculated based on property and land values.

Inheritance: A type of tax levied on individuals who inherit the estate of a deceased person.

iv. Sales tax: A consumption tax imposed by a government on the sale of goods and services. This can take

the form of a value-added tax (VAT), a goods and services tax (GST), a state or provincial sales tax, or an

excise tax.

2.9.3 Why Do We Need to Pay Taxes

Taxation has been a feature of society going back to ancient times. The role of taxes is to help governments

fund various undertakings such as public works, infrastructure, and wars. Today, taxpayer dollars are still

used for a variety of similar purposes.

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2.10 Depreciation

Depreciation is the decrease in the value of assets and the method used to reallocate the cost of a tangible

asset or physical asset (such as equipment) over its useful life span. Businesses depreciate long-term assets

for both accounting and tax purposes. The decrease in value of the asset affects the balance sheet of a

business or entity, and the method of depreciating the asset, accounting-wise, affects the net income, and thus

the income statement that they report. Generally, the cost is allocated as depreciation expense among the

periods in which the asset is expected to be used.

Any business or income-producing activity using tangible assets may incur costs related to those assets. If an

asset is expected to produce a benefit in future periods, some of these costs must be deferred rather than

treated as a current expense. The business then records depreciation expense in its financial reporting as the

current period's allocation of such costs. This is usually done in a rational and systematic manner. Generally,

this involves four criteria:

i. Cost of the asset

ii. Expected salvage value, also known as the residual value of the assets

iii. Estimated useful life of the asset

iv. A method of apportioning the cost over such life

2.10.1 Methods for Calculating depreciation

There are several methods for calculating depreciation, generally based on either the passage of time or the level of activity (or use) of the asset.

i. Straight-line depreciation

Straight-line depreciation is the simplest and most often used method. The straight-line depreciation is

calculated by dividing the difference between assets actual cost and its expected salvage value by the number

of years for its expected useful life. (The salvage value may be zero, or even negative due to costs required to

retire it; however, for depreciation purposes salvage value is not generally calculated at below zero.) The

company will then charge the same amount to depreciation each year over that period, until the value shown

for the asset has reduced from the original cost to the salvage value.

Straight-line method:

That is;

$$DE = (\text{Cost} - \text{SL}) / \text{UL}$$

For example, a vehicle that depreciates over 5 years is purchased at a cost of \$17,000, and will have a

salvage value of \$2000. Then this vehicle will depreciate at \$3,000 per year, i.e. $(17000 - 2000) / 5 = 3000$.

This table illustrates the straight-line method of depreciation. Book value at the beginning of the first

year of depreciation is the original cost of the asset. At any time book value equals original cost minus

accumulated depreciation.

Book value = original cost – accumulated depreciation

Book value at the end of year becomes book value at the beginning of next year. The asset is depreciated

until the book value equals scrap value.

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Year	Depreciation expense at year-end	Accumulated depreciation at year-end	Book value (original cost)
1	\$3,000	\$3,000	\$14,000
2	3,000	6,000	11,000
3	3,000	9,000	8,000
4	3,000	12,000	5,000
5	3,000	15,000	(scrap value)

2,000

If the vehicle was to be sold and the sales price exceeded the depreciated value (net book value) then the

excess would be considered a gain and subject to depreciation recapture. In addition, this gain above the

depreciated value would be recognized as ordinary income by the tax office. If the sales price is ever less

than the book value, the resulting capital loss is tax-deductible. If the sale price were ever more than the

original book value, then the gain above the original book value is recognized as a capital gain.

If a company chooses to depreciate an asset at a different rate from that used by the tax office then this

generates a timing difference in the income statement due to the difference (at a point in time) between

the taxation department's and company's view of the profit.

ii. Double-declining Balance Method

The double-declining-balance method is used to calculate an asset's accelerated rate of depreciation

against its non-depreciated balance during earlier years of assets useful life. When using the double-

declining-balance method, the salvage value is not considered in determining the annual depreciation, but

the book value of the asset being depreciated is never brought below its salvage value, regardless of the

method used. Depreciation ceases when either the salvage value or the end of the asset's useful life is

reached.

Double Declining Method Example

Suppose a business has bought a machine for \$ 100,000. They have estimated the machine's useful life to be

eight years, with a salvage value of \$ 11,000.

Now, as per the straight-line method of depreciation:

- Cost of the asset = \$ 100,000
- Salvage Value = \$ 11,000
- The useful life of the asset = 8 years
- Depreciation rate = $1/\text{useful life} \times 100 = (1/8) \times 100 = 12.5\%$
- Double-declining balance formula = $2 \times \text{Cost of the asset} \times \text{Depreciation rate}$.
- Here, it will be $2 \times 12.5\% = 25\%$

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- Year 1 Depreciation = $\$100,000 \times 25\% = \$25,000$

- Year 2 Depreciation = $\$75,000 \times 25\% = \$18,750$

Depreciation account of the balance sheet will look like below over the 8 years of the machine's life:

Book Value (Beginning
Year Depreciation Book Value (End Year)
Year)

1 \$1,00,000 \$25,000 \$75,000

2 \$75,000 \$18,750 \$56,250

3 \$56,250 \$14,063 \$42,188

4 \$42,188 \$10,547 \$31,461

5 \$31,461 \$7,910 \$23,730

6 \$23,730 \$5,933 \$17,798

7 \$17,798 \$4,449 \$13,348

8 \$13,348 \$2,348 \$11,000

In the above table, it can be seen that:

- In the double declining balance formula, the depreciation rate remains the same and is applied to the ending value of the last year.
- The double-declining balance depreciation value keeps decreasing over the life of the asset.
- The final double declining balance depreciation expense was \$ 2348, which is less than the actual \$3,338 (25% of \$13,348). It was done to keep the salvage value as estimated.

How to adjust the depreciation charges on the Balance sheet, Income statement, and the cash flow statement?

Let's look into how this expense is charged on the Balance sheet, income statement, and cash flow statement in detail. Let us take the double declining balance example of the machine:

1. When the machine is bought for \$ 100,000, the cash and cash equivalents are reduced by \$ 100,000 and

moved to the Property, plant, and equipment line of the balance sheet.

2. At the same time, an outflow of \$ 100,000 is shown in the cash flow statement.

3. Now, \$ 25,000 will be charged to the income statement as a depreciation expense in the first year, \$

18,750 in the second year, and so on for eight continuous years. Although all the amount is paid for the

machine at the time of purchase, the expense is charged over time.

4. Every year respective depreciation expense is added to a contra account of the balance sheet, i.e.,

Property, plant, and equipment. This is called accumulated depreciation. This is to reduce any carrying value

of the asset. Thus, after the 1 st year, the accumulated depreciation will be \$ 25000. After 2nd year it will be

\$ 43,000, and until the end of the 8th year, it will be \$ 89,000.

5. After the machine's useful life is over, the carrying value of the asset will be only \$ 11,000. The

management will sell the asset, and if it is sold above the salvage value, a profit will be booked in the income

statement or a loss if sold below the salvage value. The amount earned after selling the asset will be shown

as the cash inflow in the cash flow statement, and the same will be entered in the cash and cash equivalents

line of the balance sheet.

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The double-declining-balance method is also a better representation of how vehicles depreciate and can more

accurately match cost with benefit from asset use. The company in the future may want to allocate as little

depreciation expenses as possible to help with additional expenses. With the declining balance method, one

can find the depreciation rate that would allow exactly for full depreciation by the end of the period, using

the formula:

iii. Annuity depreciation

Annuity depreciation methods are not based on time, but on a level of Annuity. This could be miles driven

for a vehicle, or a cycle count for a machine. When the asset is acquired, its life is estimated in terms of this

level of activity. Assume the vehicle above is estimated to go 50,000 miles in its lifetime. The per-mile

depreciation rate is calculated as: $(\$17,000 \text{ cost} - \$2,000 \text{ salvage}) / 50,000 \text{ miles} = \0.30 per mile . Each year,

the depreciation expense is then calculated by multiplying the number of miles driven by the per-mile

depreciation rate.

iv. Sum-of-years-digits method

Sum-of-years-digits is a spent depreciation method that results in a more accelerated write-off than the

straight-line method, and typically also more accelerated than the declining balance method. Under this

method, the annual depreciation is determined by multiplying the depreciable cost by a schedule of fractions.

Sum of the years' digits method of depreciation is one of the accelerated depreciation techniques which are

based on the assumption that assets are generally more productive when they are new and their productivity

decreases as they become old. The formula to calculate depreciation under SYD method is:

$\text{SYD depreciation} = \text{depreciable base} \times (\text{remaining useful life} / \text{sum of the years' digits})$

$\text{depreciable base} = \text{cost} - \text{salvage value}$

Example: If an asset has original cost of \$1000, a useful life of 5 years and a salvage value of \$100,

compute its depreciation schedule.

First, determine the years' digits. Since the asset has a useful life of 5 years, the years' digits are: 5, 4, 3, 2, and 1.

Next, calculate the sum of the digits: $5+4+3+2+1=15$

The sum of the digits can also be determined by using the formula $(n^2+n)/2$ where n is equal to the useful

life of the asset in years. The example would be shown as $(5^2+5)/2=15$

Depreciation rates are as follows:

5/15 for the 1st year, 4/15 for the 2nd year, 3/15 for the 3rd year, 2/15 for the 4th year, and 1/15 for the

5th year.

Year	Depreciable base	Depreciation expense	Depreciation	Accumulated depreciation	Book value at end of year
1	900	300	300	300	600
2	900	240	540	540	360
3	900	180	720	720	180
4	900	120	840	840	60
5	900	60	900	900	0

\$1,000 (original cost)

1 900 5/15 $(900 \times 5/15)= 300$ 300 700

2 900 4/15 $(900 \times 4/15)= 240$ 540 460

3 900 3/15 $(900 \times 3/15)=180$ 720 280

4 900 2/15 $(900 \times 2/15)=120$ 840 160

5 900 1/15 $(900 \times 1/15)=60$ 900 100 (scrap value)

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v. Units-of-production depreciation method

Units-of-production depreciation method calculates greater deductions for depreciation in years when the

asset is heavily used.

Suppose an asset has original cost \$70,000, salvage value \$10,000, and is expected to produce 6,000

units.

Depreciation per unit = $(\$70,000 - \$10,000) / 6,000 = \$10$

10 × actual production (say 1,000) will give the depreciation cost of the current year to be \$10,000.

The table below illustrates the units-of-production depreciation schedule of the asset.

Units of production	Depreciation expense	Depreciation	Accumulated depreciation	Book value at year-end
1,000	10,000	10,000	10,000	60,000
1,100	11,000	21,000	21,000	49,000
1,200	12,000	33,000	33,000	37,000
1,300	13,000	46,000	46,000	24,000
1,400	14,000	60,000	60,000	10,000 (scrap value)

\$70,000 (original cost)

1,000 10 10,000 10,000 60,000

1,100 10 11,000 21,000 49,000

1,200 10 12,000 33,000 37,000

1,300 10 13,000 46,000 24,000

1,400 10 14,000 60,000 10,000 (scrap value)

Depreciation stops when book value is equal to the scrap value of the asset. In the end, the sum of accumulated depreciation and scrap value equals the original cost.

2.11 Cost of capital

Cost of capital refers to the return a company expects on a specific investment to make it worth the expenditure of resources. In other words, the cost of capital determines the rate of return required to persuade

investors to finance a capital budgeting project.

The cost of capital is heavily dependent on the type of financing used in the business. A business can be

financed through debt or equity. However, most companies employ a mixture of equity and debt financing.

Therefore, the cost of capital comes from the weighted average cost of all capital sources.

How to calculate cost of capital

To calculate the weighted average cost of capital (WACC), you must first calculate the cost of debt and the

cost of equity, which are represented by these formulas:

1. Cost of debt

The cost of debt refers to interest rates paid on any debt, such as mortgages and bonds. Interest expense is the

interest paid on current debt.

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2. Cost of equity

Cost of equity refers to the return a company requires to determine if capital requirements are met in an

investment. Cost of equity also represents the amount the market demands in exchange for owning the asset

and therefore holding the risk of ownership.

The cost of equity is approximated by the capital asset pricing model (CAPM):

- R_f = risk-free rate of return
- R_m = market rate of return
- Beta = risk estimate

3. Weighted average cost of capital

The cost of capital is based on the weighted average of the cost of debt and the cost of equity.

- E = the market value of the firm's equity
- D = the market value of the firm's debt
- V = the sum of E and D
- R_e = the cost of equity
- R_d = the cost of debt
- T_c = the income tax rate

2.11.1 Examples of cost of capital

Companies that are operating efficiently should have a cost of capital lower than or equal to their competitors

in the same industry. Here are some examples of cost of capital and how it is calculated:

Example 1

New Homes Real Estate Investment Trust is analyzing a kitchen and bathroom renovation on 25 apartment homes. The renovation will cost \$30 million and is expected to save \$5 million per year for the next five years. There is a slight risk that the renovation may not save New Homes a full \$5 million per year. New Homes could also choose to invest in a five-year bond that has the same amount of risk, with a return of 10% per year. The renovation project is expected to return 16% per year ($\$5,000,000 / \$30,000,000$). The renovation project is a better investment than the five-year bond because the required rate of return exceeds the 10% return New Homes could've gotten elsewhere.

Example 2

The newly formed Gold Company needs to raise \$1.5 million in capital to buy an office and the necessary equipment to run its business. The company raises the first \$800,000 by selling stocks. Shareholders demand a 5% return on their investment, so the cost of equity is 5%. Gold Company then sells 700 bonds for \$1,000 each to raise the remaining \$700,000 in capital. The individuals who purchase those bonds expect a 10% return, so Gold Company's cost of debt is 10%. Gold Company's total market value is \$1.5 million, and its corporate tax rate is 25%. The weighted average cost of capital can be calculated as: Gold Company's weighted average cost of capital is 6.1%.

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2.12 Inflation

Inflation is a measure of how fast prices of goods and services are rising. If inflation is occurring, leading to

higher prices for basic necessities such as food, it can have a negative impact on the overall economy.

Inflation can occur in nearly any product or service, including need-based expenses such as housing, food,

medical care, and utilities, as well as want expenses, such as cosmetics, automobiles, and jewelry. Once

inflation becomes prevalent throughout an economy, the expectation of further inflation becomes an overriding concern in the consciousness of consumers and businesses alike.

Inflation can be a concern because it makes money saved today less valuable tomorrow. Inflation erodes a

consumer's purchasing power and can even interfere with the ability to retire. For example, if an investor

earned 5% from investments in stocks and bonds, but the inflation rate was 3%, the investor only earned 2%

in real terms. In this article, we'll examine the fundamental factors behind inflation, different types of

inflation, and who benefits from it.

2.12.1 What Drives Inflation

There are various factors that can drive prices or inflation in an economy. Typically, inflation results from

an increase in production costs or an increase in demand for products and services.

i. Cost-Push Inflation

Cost-push inflation occurs when prices rise because production costs increase, such as raw materials and

wages. The demand for goods is unchanged while the supply of goods declines due to the higher costs of

production. As a result, the added costs of production are passed onto consumers in the form of higher

prices for the finished goods.

One of the signs of possible cost-push inflation can be seen in rising commodity prices such as oil and

metals since they're major production inputs. For example, if the price of copper rises, companies that use

copper to make their products might increase the prices of their goods. If the demand for the product is

independent of the demand for copper, the business will pass on the higher costs of raw materials to

consumers. The result is higher prices for consumers without any change in demand for the products

consumed.

Wages also affect the cost of production and are typically the single biggest expense for businesses. When

the economy is performing well, and the unemployment rate is low, shortages in labor or workers can occur.

Companies, in turn, increase wages to attract qualified candidates, causing production costs to rise for the

company. If the company raises prices due to the rise in employee wages, cost-plus inflation occurs.

Natural disasters can also drive prices higher. For example, if a hurricane destroys a crop such as corn,

prices can rise across the economy since corn is used in many products.

ii. Demand-Pull Inflation

Demand-pull inflation can be caused by strong consumer demand for a product or service. When there's a

surge in demand for a wide breadth of goods across an economy, their prices tend to increase.

While this is

not often a concern for short-term imbalances of supply and demand, sustained demand can reverberate in

the economy and raise costs for other goods; the result is demand-pull inflation.

Consumer confidence tends to be high when unemployment is low, and wages are rising—leading to more

spending. Economic expansion has a direct impact on the level of consumer spending in an economy, which

can lead to high demand for products and services.

As the demand for a particular good or service increases, the available supply decreases. When fewer items

are available, consumers are willing to pay more to obtain the item—as outlined in the economic principle

of supply and demand. The result is higher prices due to demand-pull inflation.

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Companies also play a role in inflation, especially if they manufacture popular products. A company can

raise prices simply because consumers are willing to pay the increased amount. Corporations also raise

prices freely when the item for sale is something consumers need for everyday existence, such as oil and

gas. However, it's the demand from consumers that provide corporations with the leverage to raise prices.

iii. Built-In Inflation and Rising Wages

Built-in inflation occurs when enough people expect inflation to continue in the future. As the price of

goods and services rises, people may come to believe in a continuous rise in the future at a similar rate.

Because of these shared expectations, workers may start to demand higher wages in order to anticipate

rising prices and maintain their standard of living. Increased wages would result in higher costs for businesses, which may pass those costs on to consumers. Higher wages also increase consumers' disposable

income, increasing the demand for goods that can push prices even higher. A wage-price spiral can then be

set in place as one factor feeds back into the other and vice-versa.

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MODULE 3

SENSITIVITY ANALYSIS AND RISK ASSESSMENT

Sensitivity analysis and risk assessment are essential tools for decision analysis, especially when dealing with

uncertainty and complexity. They help you evaluate how different factors affect your outcomes, identify the

most critical sources of risk, and compare alternative scenarios.

3.1 How Sensitivity Analysis Works

Sensitivity analysis is a financial model that determines how target variables are affected based on changes

in other variables known as input variables. It is a way to predict the outcome of a decision given a certain

range of variables. By creating a given set of variables, an analyst can determine how changes in one

variable affect the outcome.

Sensitivity analysis can be used to help make predictions about the share prices of public companies. Some

of the variables that affect stock prices include company earnings, the number of shares outstanding,

the debt-to-equity ratios (D/E), and the number of competitors in the industry.

3.2 Usefulness of Sensitivity Analysis

Financial models that incorporate sensitivity analysis can provide management a range of feedback that is

useful in many different scenarios. The breadth of the usefulness of sensitivity analysis includes but is not

limited to:

- i. Understanding influencing factors. This includes what and how different external factors interact with a specific project or undertaking. This allows management to better understand what input variables may impact output variables.
 - ii. Reducing uncertainty. Complex sensitivity analysis models educate users on different elements impacting a project; this in turn informs members on the project what to be alert for or what to plan in advance for.
 - iii. Catching errors. The original assumptions for the baseline analysis may have had some uncaught errors. By performing different analytical iterations, management may catch mistakes in the original analysis.
 - iv. Simplifying the model. Overly complex models may make it hard to analyze the inputs. By performing sensitivity analysis, users can better understand what factors don't actually matter and can be removed from the model due to its lack of materiality.
 - v. Communicating results. Upper management may already be defensive or inquisitive about an undertaking. Compiling analysis on different situations helps inform decision-makers of other outcomes they may be interested in knowing about.
 - vi. Achieving goals. Management may lay long-term strategic plans that must meet specific benchmarks. By performing sensitivity analysis, a company can better understand how a project may change and what conditions must be present for the team to meet its metric targets.
- Because sensitivity analysis answers questions such as "What if XYZ happens?", this type of analysis is also called what-if analysis.

3.3 Sensitivity vs. Scenario Analysis

In finance, a sensitivity analysis is created to understand the impact a range of variables has on a given outcome. It is important to note that a sensitivity analysis is not the same as a scenario analysis. As an example, assume an equity analyst wants to do a sensitivity analysis and a scenario analysis around the impact of earnings per share (EPS) on a company's relative valuation by using the price-to-earnings (P/E) multiple.

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The sensitivity analysis is based on the variables that affect valuation, which a financial model can depict using the variables' price and EPS. The sensitivity analysis isolates these variables and then records the range of possible outcomes.

3.4 Risk Analysis

The term risk analysis refers to the assessment process that identifies the potential for any adverse events that may negatively affect organizations and the environment. Risk analysis is commonly performed

by corporations (banks, construction groups, health care, etc.), governments, and nonprofits.

Conducting a

risk analysis can help organizations determine whether they should undertake a project or approve a

financial application, and what actions they may need to take to protect their interests. This type of analysis

facilitates a balance between risks and risk reduction. Risk analysts often work in with forecasting professionals to minimize future negative unforeseen effects.

Note that;

- i. Risk analysis seeks to identify, measure, and mitigate various risk exposures or hazards facing a business, investment, or project.
- ii. Quantitative risk analysis uses mathematical models and simulations to assign numerical values to risk.
- iii. Qualitative risk analysis relies on a person's subjective judgment to build a theoretical model of risk for a given scenario.
- iv. Risk analysis can include risk benefit, needs assessment, or root cause analysis.
- v. Risk analysis entails identifying risk, defining uncertainty, completing analysis models, and implementing solutions.

3.5 Understanding Risk Analysis

Risk assessment enables corporations, governments, and investors to assess the probability that an adverse

event might negatively impact a business, economy, project, or investment. Assessing risk is essential for

determining how worthwhile a specific project or investment is and the best process(es) to mitigate those

risks. Risk analysis provides different approaches that can be used to assess the risk and reward tradeoff of a

potential investment opportunity. A risk analyst starts by identifying what could potentially go wrong.

These negatives must be weighed against a probability metric that measures the likelihood of the event

occurring. Finally, risk analysis attempts to estimate the extent of the impact that will be made if the event

happens. Many risks that are identified, such as market risk, credit risk, currency risk, and so on, can be

reduced through hedging or by purchasing insurance.

3.6 Replacement Decision

Decision regarding replacement of an existing asset with another is based on the net present value and internal rate of return of the incremental cash flows, i.e. the difference between periodic net cash

flows if the existing asset is kept and the periodic net cash flows if the asset is replaced.

In capital budgeting and engineering economics, the existing asset is called the defender and the asset which

is proposed to replace the defender is called the challenger. Estimation of incremental cash flows for such

replacement analysis involves calculation of net cash flows of the defender, net cash flows of the challenger

and then finding the difference in cash flows for both the assets.

Calculating periodic cash flows of existing asset is straight forward. Since the existing asset is already

purchased, the initial investment outlay is zero and the periodic net cash flows are calculated based on the

following formula:

Net cash flows = (revenue - operating expenses - depreciation) x (1 - tax rate) + depreciation

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If the asset is replaced, it involves investment in the new asset and sale or disposal of the existing asset.

Disposal of existing asset has some income tax implications which need to be reflected in the calculation of

initial investment as follows:

Initial investment after replacement = cost of new asset - sale proceeds of old asset +/- tax on disposal asset

= (sale proceeds of old assets - book value of old asset) * tax rate

As evident from the equation above, if the old asset is sold at an amount higher than its book value, the

company bears a related tax cost which is added to the initial investment. Similarly, if the sale proceeds are

lower than the book value of the asset sold, there is a resulting tax shield which is subtracted from sum of

cost of new asset and sale proceeds of the old asset.

Example: TIBUS operates buses on different inter-city routes. The management is considering upgrading its

fleet of 50 standard buses (purchased at \$8 million) which has a book value of \$3 million. It expects to sell

the existing fleet for \$4 million and purchase a new fleet at a cost of \$12 million. The existing revenue of the

fleet is \$4 million per annum which is expected to rise by 25% per annum if the new fleet is introduced. The

existing operating cost of the fleet is \$2 million which is expected to drop by 30% after up-gradation.

Determine if replacement is a good idea if the company's weighted average cost of capital is 10% and the

analysis period is 8 years. The company pays taxes at the rate of 33% and it charges depreciation on straight

line basis.

Solution: In analysing whether to replace the fleet or not, we need to work out the net cash flows of the

existing fleet and net cash flows after the replacement and then finding the difference between both to arrive

at the incremental cash flows. The incremental cash flows are then used to calculate net present value and/or

internal rate of return.

Existing fleet

Capital investment = 0

Revenue = \$4 million

Operating cost = \$2 million

Annual depreciation = \$8 million/8 = \$1 million

Periodic net cash flows = (revenue - operating cost - depreciation) * (1 - tax rate) + depreciation

Periodic net cash flows = (\$4 million - \$2 million - \$1 million) * (1 - 33%) + \$1 million = \$1.67 million

Replaced fleet

Cost of new fleet = \$12 million

Sale proceeds of old fleet = \$4 million

Book value of old fleet = \$3 million

Tax on sale of old fleet = (\$4 million - \$3 million) * 33% = \$0.33 million

Net initial investment = \$12 million - \$4 million + \$0.33 million = \$8.33 million

New revenue = \$4 million * 1.25 = \$5 million

New operating cost = \$2 million * (1 - 10%) = \$1.8 million

Depreciation expense on new fleet = \$12 million/8 = \$1.5 million

Periodic net cash flows = (\$5 million - \$1.8 million - \$1.5 million) * (1 - 33%) + \$1.5 million = \$2.64 million

Incremental cash flows

Incremental cash flows at time 0 = net investment after replacement - net investment before replacement

Incremental cash flows at time 0 = \$8.33 million - 0 = \$8.33 million

Incremental cash flows in Year 1 to 8 = net annual cash flows after replacement - net annual cash flows before replacement

Incremental cash flows in Year 1 to 8 = \$2.64 million - \$1.67 million = \$0.97 million

Calculating net present value and IRR:

NPV = PV factor for 8 years at 10% * incremental cash flows (Year 1-8) - incremental initial investment

NPV = 5.335 * \$0.97 million - \$8.33 million

NPV = -\$3.15 million

IRR = -2%

Since the net present value is negative and IRR is way below the hurdle rate, it is not feasible to replace the fleet at this stage.

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MODULE 4

ENGINEERING AND PUBLIC SECTOR DECISION MAKING

4.1 Decision-making

Decision-making is commonly thought of as the cognitive process one undergoes before choosing a particular course of action. Engineers are trained to approach problems from a relatively more pragmatic and

logical perspective. Engineers value efficiency, utility, safety and minimizing cost over aesthetics and flair.

Decision making involves defining clearly the objectives of solving a specific problem, generate all possible

solutions, predict the outcome of each solution and determine the best solution by balancing the pros and

cons along with cost and benefit

A decision is a commitment to use resources. Decision-making is one of the most fundamental yet complex

psychological processes performed regularly by human beings. Because individuals perform a variety of

tasks and therefore execute several different types of decisions every day, formulating an exact definition for

decision making can be difficult. According to Jonassen, a decision represents an ill-structured problem

where one has to evaluate multiple options and commit to one of them. There are two different ways to

understand how decisions are made. From a rational perspective, decisions are made to maximize utility.

From a naturalistic perspective, decisions are made with less practicality but more influence from personal

beliefs and prior experiences. Decisions are made under three types of circumstances, namely:

i. Risk: the information is unavailable, but probabilistic models can be used because the distributions of

random variables are known.

ii. Uncertainty: the probability distributions are unavailable, but other obstacles are known (such as radiation affecting transmission from an antenna)

iii. Ambiguity: functional form is unknown, and trial-and-error testing may be needed even to determine

inputs and outputs

All decisions can be sorted into one of four categories:

i. Choices: selection of a subset from a larger set of alternatives

ii. Acceptances/rejections: the binary decision

iii. Evaluations: assigning worth to an option

iv. Constructions: attempting to create an ideal solution given available resources

4.2 Decision Theory

Several decision models have been created to shed light how an individual commits to a decision.

i. Normative models assume that individuals are rational and seek to optimize resources.

ii. Rational choice models: the individual identifies a set of options, determines the criteria (usually quantitative) for evaluating the options, weighs each option, and selects the option with the highest score

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iii. Cost-benefit analysis: used by corporations and governments, this type of decision-making is used

when weighing business decisions and new policies. Attempts to quantify values associated with each

decision

iv. Risk assessment models: using probability to analyze games of chance. Evaluating expected values

while considering the consequences of both false positives and false negatives

An example of a risk assessment model is when a patient or doctor has to make a medical decision. For

example, an ill patient may have to weigh the risk and benefits of the two following treatments: Treatment A

has a projected 20% chance of death and 80% chance of 35 years of normal life after the treatment.

Treatment B has a 100% chance of survival with a certainty of 18 years of normal life. Naturalistic models

are based on research or results obtained from actual humans. These models are influenced heavily by emotions and experiences.

The normative model: assume that individuals make decisions based on the concept of unbounded rationality, which is impossible in practicality. Individuals also unconsciously gauge the benefit of decisions

based on values and probabilities inherent to a situation. In other words, individuals are vulnerable to errors

in reasoning, judgmental biases, representational faults, and coping defects. For example, an individual often

believes that because he/she has already invested so much in making something happen, he/she cannot turn

back before the task is complete.

Narrative-based decision-making: making decisions based on explanations provided by other unreliable

sources (testimonies during trials).

Identity-based decision-making: making decisions consistent with one's image of oneself. An individual

may ask him/herself: What does a person like me do?

4.3 Decision-Making Techniques

Smart decision-making is essential when making decisions on a managerial level that may affect entire

companies or even countries. Though humans cannot behave perfectly rationally, several techniques have

been developed to minimize resources and maximize benefits.

Decision matrices: list all practical options in rows. Create a set of criteria (qualitative and quantitative) and

use them as the columns of your matrix. When creating a decision matrix, the individual essentially draws

out a visual representation of the pros and cons of each option.

SWOT (strengths, weaknesses, opportunities, threats) analysis: when attempting to make a managerial

decision, this model allows the individual to analyze both the internal and external forces at work. Internal

forces are the strengths and weaknesses of the company, for example, and external forces are the opportunities and threats. After creating a list of variables central to the problem at hand, break them down

into these four categories and quantize them.

Force field analysis: examine the forces for and against a certain action.

Argumentation: as a project manager, it may be helpful to have different people argue for different options.

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4.4 Case Study

A firm that manufactures paper products considers a new project that replaces part of the electricity

purchased from the local electricity company, with their own electricity produced from solar energy. The

firm works 24 hours/day, 7 days/week, and consumes 5 MW/hour. The management is considering a 2.5

MWP solar system that would produce half of the daytime electricity consumption, interacting with the

utility power grid. We assume that yearly cash flow savings will depend on the cost of purchased electricity

and that the efficiency of the system is reduced by 0.3% per year.

Let net present value (NPV) = sum of the present values of the individual cash flows of the same entity, $CF =$

Cash Flow, CF_0 = Initial outlay for the project, CF_1 = Cash received at the end of year 1, and I = discounting

rate of cash flow. Scenario A – If the cost of electricity (daytime rate) will climb to 0.15\$/KWH, with a

probability of 30%, the estimated yearly savings will rise to \$1,938,000. NPV scenario A:

$CF_0 = -\$12,053,000;$

$CF_1 = \$1,938,000;$

The frequency of cash received in 1 year is repeated for 20 years.

$I = 8.2\%;$

$NPV = \$6,694,682.$

Scenario B – with the present cost of electricity (daytime rate) of 0.10 \$/KWH, estimated savings are some

\$1,212,000/year. Probability of scenario B is 40%. NPV scenario B:

$CF_0 = -\$12,053,000;$

$CF_1 = \$1,212,000;$

The frequency of cash received in 1 year is repeated for 20 years.

$I = 8.2\%;$

$NPV = -\$328,443.$

Scenario C – If the cost of electricity will decline to 0.05 \$/KWH, with a probability of 30%, the estimated

yearly savings will drop to \$486,000. NPV scenario C:

$CF_0 = -\$12,053,000;$

$CF_1 = \$486,000;$

The frequency of cash received in 1 year is repeated for 20 years.

$I = 8.2\%$;

$NPV = -\$7,351,569$.

If we proceed with the investment today, we can calculate the Expected Net Present Value (NPV) as follows:

$E[NPV] = 0.30(\$6,694,682) + 0.40(-\$328,443) + 0.30(-\$7,351,569) = -\$328,443$

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The conclusion is that if we invest in the project today, we can expect a negative NPV, which will lead us to

reject the project.

Application to Project: To illustrate the diversity of decision-making, one can analyze the logic behind

decisions made for a Tufts University electrical engineering senior design project. As three ambitious young

engineering students with limited time in the semester, Team Blue could not afford to waste project sessions

doing less meaningful tasks. They were attempting to create a noise-canceling iPhone application that

predicts and cancels low-frequency ambient noise using only an iPhone and the default Apple EarPods

headset. Because raw materials and employee salaries were non-factors, Team Blue's most valuable resource

was time. A crucial part of the semester was choosing tasks to devote time to, which was a decision-making

process. First, the team had to decide what they wanted to accomplish by the end of the semester. With the

guidance of their advisor, they scaled down the project as much as possible within reason level and focused

on learning new competencies, such as digital signal processing, controls, communications, and probability

concepts along with how to build iPhone applications. To maximize their human resources, they decided that

at least two people must be proficient with each competency in the event that one of them could not be

present for a project session. After weighing in each person's strengths and weaknesses, individuals were

assigned to work on an aspect of the project that the team thought was most suitable given his background.

The result at the end of the semester was a working MATLAB algorithm and the skeleton of an iPhone

application with basic recording and playback functionality. Though Team Blue fell short of their initial goal,

the decisions they made allowed them to lay the foundation for the work they will do next semester.

In conclusion, engineer makes decisions for a living by creating new requirements, selecting design concepts, choosing components, and optimizing parameters. The engineer bridges ideas and reality through several iterations of decision-making, and each decision can be thought of as a marker of the design from initiation to implementation to termination. And design is “a decision making process (often iterative) in which basic sciences, mathematics, and the engineering sciences are applied to convert resources optimally to meet stated needs. Roth warns that decision-making should not simply be a number-crunching exercise. For an engineer to make truly great decisions, he/she must also pursue innovative thinking, idea management, and corporate learning.

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MODULE 5

ENGINEERING BUSINESS

An engineering services company is a project-based business that provides specialized consulting services to

help clients bring products or technology to market. And this required certain organizational structure for effective coordination

5.1 Organizational Structure

An organizational structure is a system that outlines how certain activities are directed in order to achieve

the goals of an organization. These activities can include rules, roles, and responsibilities.

The organizational structure also determines how information flows between levels within the company. For

example, in a centralized structure, decisions flow from the top down, while in a decentralized structure,

decision-making power is distributed among various levels of the organization. Having an organizational

structure in place allows companies to remain efficient and focused. Not having a formal structure in place

may prove difficult for certain organizations. For instance, employees may have difficulty knowing to

whom they should report. That can lead to uncertainty as to who is responsible for what in the organization.

Having a structure in place can help with efficiency and provide clarity for everyone at every level. That

also means each and every department can be more productive, as they are likely to be more focused on

energy and time.

5.1 Types of Organizational Structures

i. Functional Structure

Four types of common organizational structures are implemented in the real world. The first and most

common is a functional structure. This is also referred to as a bureaucratic organizational structure and

breaks up a company based on the specialization of its workforce. Most small-to-medium-sized businesses

operations is the act of using a bureaucratic organizational structure.

ii. Divisional or Multidivisional Structure

The second type is common among large companies with many business units. Called the divisional or

multidivisional (M-Form) structure, a company that uses this method structures its leadership team based on

the products, projects, or subsidiaries they operate. A good example of this structure is Johnson & Johnson.

With thousands of products and lines of business, the company structures itself so each business unit

operates as its own company with its own president.

Divisions may also be designated geographically in addition to specialization. For instance, a global corporation may have a North American Division and a European Division.

iii. Team-Based

Similar to divisional or functional structures, team-based organizations segregate into close-knit teams of

employees that serve particular goals and functions, but where each team is a unit that contains both leaders

and workers.

iv. Flat (Flatarchy) Structure

Flatarchy, also known as a horizontal structure, is relatively newer, and is used among many startups. As the

name alludes, it flattens the hierarchy and chain of command and gives its employees a lot of autonomy.

Companies that use this type of structure have a high speed of implementation.

v. Matrix Structure

Firms can also have a matrix structure. It is also the most confusing and the least used. This structure

matrixes employees across different superiors, divisions, or departments. An employee working for a

matrixed company, for example, may have duties in both sales and customer service.

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vi. Circular Structure

Circular structures are hierarchical, but they are said to be circular as it places higher-level employees and managers at the center of the organization with concentric rings expanding outward, which contain lower-level employees and staff. This way of organizing is intended to encourage open communication and collaboration among the different ranks.

vii. Network Structure

The network structure organizes contractors and third-party vendors to carry out certain key functions. It features a relatively small headquarters with geographically-dispersed satellite offices, along with key functions outsourced to other firms and consultants.

5.2 Benefits of Organizational Structures

Putting an organizational structure in place can be very beneficial to a company. The structure not only defines a company's hierarchy but also allows the firm to lay out the pay structure for its employees. By putting the organizational structure in place, the firm can decide salary grades and ranges for each position. The structure also makes operations more efficient and much more effective. By separating employees and functions into different departments, the company can perform different operations at once seamlessly. In addition, a very clear organizational structure informs employees on how best to get their jobs done. For example, in a hierarchical organization, employees will have to work harder at buying favor or courting those with decision-making power. In a decentralized organization, employees must take on more initiative and bring creative problem solving to the table. This can also help set expectations for how employees can track their own growth within a company and emphasize a certain set of skills—as well as for potential employees to gauge if such a company would be a good fit with their own interests and work styles.

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MODULE 6

MANAGEMENT OF ENGINEER TO MANAGE TRANSITION

6.1 Research and development (R&D)

Research and development (R&D) is a valuable tool for growing and improving your business. R&D involves researching your market and your customer needs and developing new and improved products and

services to fit these needs. Businesses that have an R&D strategy have a greater chance of success than

businesses that don't. An R&D strategy can lead to innovation and increased productivity and can boost your

business's competitive advantage.

6.2 Developing an R&D strategy

Your R&D strategy depends on the size of your business. In small businesses, R&D tends to focus more on

product improvement because of budget and cost limitations. Larger businesses may be able to dedicate more

time and resources to R&D to introduce new products as well as improve existing ones. The benefits of R&D

are often long-term, so it's important to remember that your investment in it may not result in short-term

profits. As well as product development and improvement, R&D can help you develop more efficient

processes and new ways of delivering services.

Spending more time and money on R&D does not guarantee it will be successful. The key to successful

R&D is extensive market research to identify the needs and desires of your customers. You will need to

revisit this research regularly as customer preferences frequently change.

6.3 Innovation in your business

R&D can lead to innovations in your business. These may be in terms of new products and services, improved processes and new ways to interact with your customers. These innovations can result in greater

profits and lower costs. Innovation is also a useful way to grow your business.

6.4 Support for R&D

Once new and improved products have been developed through R&D, businesses often seek to commercialise them so they can be sold on the market.

What is project selection?

Project selection is the evaluation of project ideas to help decide which project has the highest priority. It's an

important part of project portfolio management (PPM), which is a process used by project management

organizations (PMOs) and project managers to analyze the potential return on undertaking a project.

Once project managers receive project ideas or proposals, they often go through a process to assess and select

a project that will move forward. Typically, when project managers select a project, they may consider the

following factors:

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- i. Costs
- ii. Resources
- iii. Benefits or Return on Investment (ROI)
- iv. Time to complete the project
- v. Risks associated with the project

6.5 Why is project selection important

Project selection is important because companies want to make sure projects they invest in are safe and will

yield benefits and good returns. The process of project selection can analyze new opportunities and help

justify the decisions for making needed monetary investments. Companies may have several project

opportunities to invest in, but because they can't invest in all projects, they are often selective.

How to select a project

Here are some steps you can take to select a project:

- i. Make sure the project fits the company's strategy. Discuss with all stakeholders whether a project you're considering fits into the company's overall business strategy. You can work together to identify

where a project may meet a single organizational goal or multiple goals, and whether they are short-

term or long-term goals.

- ii. Understand your company environment. It's important to be you're aware of your organizational environment and understand your company thoroughly. Consider asking yourself the following questions:

- What are the company's key business drivers?
- What are the company's strengths and weaknesses?
- Does the company have limited resources?
- If the company has resource limitations, where is it lacking?

- iii. Consider and analyze historical data. When you perform your analysis, consider previous experiences

and refer to past information or historical data as much as possible. Regardless of the outcome of a prior project, there are environmental and organizational factors that likely influenced the outcome. Find out whether there are changes to those factors, and discuss them with company executives or project stakeholders.

- iv. Decide who will be the project champion. It's important to ensure a project has a designed champion

or owner to make sure the project stays on track and proceeds smoothly and efficiently to completion.

The project champion is often a high-level employee or executive with an interest in the project and excellent communication skills to work with everyone involved with the project.

6.6 Project selection methods

There are several methods you can use to help you decide which project to select. These include:

a. Cost-benefit analysis: Before you take on a new project, you can perform a cost-benefit analysis to

assess all the potential costs and income that your organization might generate from the project.

The

result of the analysis may uncover whether the project is workable or if the company should consider

another project.

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b. Payback period: Another method you can use is the payback period, where you determine the time

to recover the cost of an investment. For instance, if you project you will spend \$300,000 to execute a

project and expect it will generate revenue at the rate of \$30,000 per year, your payback period would

be 10 years.

c. Discounted cash flow: A discounted cash flow (DCF) analysis allows you to estimate the money your company would receive from an investment or project, adjusted for the time value of money.

The time value of money presumes that a dollar today is worth more than a dollar tomorrow because

you have invested it. For instance, with a 5% annual interest rate, \$1 in a savings account will be worth \$2.12 after a year. Likewise, if you delay a \$1 payment for a year, its present value is 95 cents,

as you can't transfer it to your savings account to earn interest. In order to perform a DCF analysis, it's important for a company to:

- Make estimates about future cash flows
- Make estimates about the ending value of the project
- Determine an appropriate discount rate for the DCF model

d. Opportunity costs: Opportunity costs refer to alternative benefits the company may realize when choosing one alternative over another. Putting opportunity costs into consideration allows you to weigh the

benefits from alternative courses of action and not just the current choice or path being considered in the

cost-benefit analysis. By factoring in all options and the potential missed opportunities, a company's cost-

benefit analysis is more in-depth and allows for better decision-making.

6.6.1 Ranking method

Ranking is a simple approach that ranks the projects on a scale according to their importance. The benefit of

the ranking method is its quick approach that enables you to identify top priorities. It works effectively when

there are limited criteria to assess and it's easy to evaluate the factors involved. Before you assign rankings,

it's important to ask yourself questions such as:

- What's the rate of return for this project?
- What are the qualitative and quantitative benefits to the stakeholders?
- How do efficiencies improve if the company executes this project?
- Does the company have the capability and capacity to execute the project?

6.6.2 Scoring model

The scoring model works when there are several selection criteria to consider and projects being compared

are very different. Instead of choosing one or two criteria, the scoring model considers one to two groups of

factors, such as risk, return on investment (ROI), benefits and strategic alignment. Besides assigning a rating

to each criterion, you give weight to each group. For example, the risk may have a factor of 0.75, whereas

benefits may have 1.5. You then compute the weighted score to arrive at the final project score.

6.6.3 Analytic hierarchy process

The analytic hierarchy process (AHP) combines subjective elements with mathematical models to give a

more holistic approach than other methods. The AHP method works with several selection criteria by pitting

every two criteria against each other, which decreases the likelihood of biases and errors. After this "apples

to apples" type of comparison, you can normalize the values and compute the weighted score. The AHP's

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strong reliance on quantitative strategies is its strength, as it converts an abstract problem into numbers and

gives transparency to the reason behind your decision.

6.7 Project management

Project management is the application of processes, methods, skills, knowledge and experience to achieve

specific project objectives according to the project acceptance criteria within agreed parameters. Project

management has final deliverables that are constrained to a finite timescale and budget.

A key factor that distinguishes project management from just 'management' is that it has this final deliverable

and a finite timespan, unlike management which is an ongoing process. Because of this a project professional

needs a wide range of skills; often technical skills, and certainly people management skills and good business

awareness.

6.7.1 Stages of a project management

A project management life cycle consists of 5 distinct phases including initiation, planning, execution, monitoring, and closure that combine to turn a project idea into a working product.

That is;

- a. Project Initiation.
- b. Project Planning.
- c. Project Execution.
- d. Project Monitoring and Controlling.
- e. Project Closing

a. Phase 2: Project initiation

The project initiation phase is the first stage of turning an abstract idea into a meaningful goal. In this stage, you need to

develop a business case and define the project on a broad level. In order to do that, you have to determine the need for

the project and create a project charter. The project charter is an important document consisting of details like the project

constraints, goals, appointment of the project manager, budget, expected timeline, etc.

Once you have the project goals and project scope, identify key project stakeholders—the people who are to be involved

in the project. Create a stakeholder register with the roles, designation, communication requirements, and influence.

While a clear goal of the project is established in this phase, a project charter does not contain any technical details that

happen in the planning stage. Consider the example of an automobile manufacturer assigned to develop an electric

vehicle. The selection of the design, capacity, and battery power of the vehicle will not be a part of the initiation phase.

The only certainty would be that an electric vehicle will be developed within the given timeframe and budget.

b. Phase 2: Project planning

The project planning stage requires complete diligence as it lays out the project's roadmap. Unless you are using a

modern project management methodology like agile project management, the second phase of project management is

expected to take almost half of the entire project's timespan. In this phase, the primary tasks are identifying technical

requirements, developing a detailed project schedule, creating a communication plan, and setting up goals/deliverables.

There are several methods of setting up the project's goals but S.M.A.R.T. and C.L.E.A.R. are the most popular.

S.M.A.R.T Goals: The 'SMART' criteria ensure that the goals you set for your project are critically analyzed. It is an

established method that reduces risk and allows project managers to make clearly defined and achievable goals.

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The acronym SMART stands for

C.L.E.A.R. Goals:

The 'CLEAR' method of setting up goals is designed to cater to the dynamic nature of a modern workplace. Today's

fast-paced businesses require flexibility and immediate results and CLEAR can help citizen developers with that. The

acronym for CLEAR stands for;

During the planning stage, the scope of the project is defined. There is a possibility of changing the scope of the project

demands it but the project manager must approve the change. Project managers also develop a work breakdown

structure (WBS), which clearly visualizes the entire project in different sections for the team management. A

detailed project timeline with each deliverable is another important element of the planning stage. Using that timeline,

project managers can develop a project communication plan and a schedule of communication with the relevant

stakeholders.

Risk mitigation is another important aspect of project management that is a part of the planning stage. The project

manager is responsible for extrapolating past data to identify potential project management risks and develop a strategy

to minimize them. An important element that professionals often overlook is an effective change management plan. As

a project manager, you must be ready to incorporate a few changes in the project to avoid bottlenecks and project delays.

In the absence of a working change management plan, scope creep happens and causes huge problems for the project

team in the later stages of the project. So, it's best to reduce the possibility of unforeseen changes as much as possible

c. Phase 3: Project execution

The project execution stage is where your team does the actual work. As a project manager, your job is to

establish efficient workflows and carefully monitor the progress of your team. Another responsibility of the project

manager during this phase is to consistently maintain effective collaboration between project stakeholders. This ensures

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that everyone stays on the same page and the project runs smoothly without any issues. You can take help from the

best project collaboration tools that are available in the market. They'll not only make it easier but also improve

efficiency and increase the productivity of your team.

d. Phase 4: Project monitoring and controlling

In the project management process, the third and fourth phases are not sequential in nature. The project monitoring and

controlling phase run simultaneously with project execution, thereby ensuring that objectives and project

deliverables are met. As a project manager, you can make sure that no one deviates from the original plan by

establishing Critical Success Factors (CSF) and Key Performance Indicators (KPI). During the monitoring phase of

project management, the manager is also responsible for quantitatively tracking the effort and cost during the process.

This tracking not only ensures that the project remains within the budget but also is important for future projects.

e. Phase 5: Project closing

This is the final phase of the project management process. The project closure stage indicates the end of the project after

the final delivery. There are times when external talent is hired specifically for the project on contract. Terminating these

contracts and completing the necessary paperwork is also the responsibility of the project manager. Most teams hold a

reflection meeting after the completion of the project in order to contemplate their successes and failures during the

project. This is an effective method to ensure continuous improvement within the company to enhance the overall

productivity of the team in the future. The final task of this phase is to review the entire project complete a detailed

report that covers every aspect. All of the necessary data is stored in a secure place that can be accessed by project

managers of that organization.

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